Ledgers of the Self-Employed: Accounting for the Invisible Firm

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Abstract

Standard household surveys report positive profits for virtually every self-employed worker in the developing world. Using a seven-day financial diary with full double-entry accounting administered to 300 informal firms across four Bolivian cities—14,902 individual transactions—we construct complete income statements, balance sheets, and cash-flow statements and compare them against matched responses from the same firms in Bolivia's *Encuesta de Hogares*. The results are at odds with the survey picture in ways that bear on the evidentiary foundations of the informal-sector policy literature.

Once owner and family labour are valued at the statutory minimum wage, depreciation is charged on productive assets, and informal debt service is correctly identified, 57 percent of firms are loss-making. The median accounting gap between survey-implied and diary-adjusted profit equals Bs 1,191 per week—121 percent of the conventional figure—driven overwhelmingly by unpaid labour (88 percent of the gap). The bias is not unidirectional: cost omission inflates profits while income under-declaration partially offsets it, and the net direction varies by sector and debt exposure in ways that preclude correction by a scalar adjustment factor. Among the minority of firms that remain viable after full cost accounting, a going-concern valuation at the informal-sector discount rate yields a median present value 83 times book value, quantifying the location rents, client relationships, and operational knowledge that conventional accounting cannot see and that formalisation policy implicitly attempts to unlock.

These measurement facts carry consequences for how the literature has interpreted the informal sector. The Lewis (1954) concept of disguised unemployment—workers whose marginal product falls below the subsistence wage—has resisted empirical operationalisation for seven decades because it requires knowing the true cost of the worker's own time. The diary methodology supplies this: we formalise disguised unemployment as a computable condition on the adjusted income statement and show that 57 percent of our sample satisfies it. The randomised experiments documenting high returns to capital in microenterprises—De Mel et al. (2008), McKenzie and Woodruff (2008), Fafchamps et al. (2014)—estimate average treatment effects across a population that is majority disguised-unemployed; the high averages reflect the entrepreneurial minority. And the microcredit industry, which deploys tens of billions of dollars annually on the premise that informal self-employed workers are credit-constrained entrepreneurs, is in our data extending loans at 201 percent effective annual interest to workers already earning negative adjusted returns on their time. A systematic adequacy scorecard finds that 13 of 24 financial concepts required for a complete firm account are absent from Bolivia's household survey, including all four items that determine which side of the loss threshold a firm falls on.

Keywords: informal firms, financial diaries, accounting, disguised unemployment, measurement error, household surveys, Bolivia, going-concern valuation

The opinions expressed in this paper do not reflect the position of any affiliated institution. All errors are the author's.

1. Introduction

In 2023 the International Labour Organization estimated that 2 billion workers—61 percent of the global employed population—work in the informal economy (ILO, 2023). The World Bank's *World Development Report 2019* placed informality at the centre of the inclusive growth agenda (World Bank, 2019). The IDB characterises the region's informal self-employed as entrepreneurs constrained by regulation and credit, for whom formalisation and microfinance are the twin instruments of welfare improvement (IDB, 2020). ECLAC classifies informal self-employment as occupied in every member country's poverty statistics, on the grounds that any occupation raises income above subsistence (ECLAC, 2023). These are the operational assumptions of the largest development institutions in the world, encoded into poverty lines, transfer eligibility criteria, and investment priorities affecting hundreds of millions of people.

Every one of these assumptions rests on the self-employment income module of national household surveys. In Bolivia, as in most developing countries, this module asks the respondent to recall annual gross revenue and eight categories of costs and to report the net income kept for the household. The resulting figure classifies the worker as non-poor, productive, and entrepreneurially active. The present paper uses seven days of transaction-level data from 300 Bolivian informal firms to show that this figure is wrong—not marginally biased in one direction, but reversed in sign for the majority of workers.

The source of the error is elementary. A firm's profit is revenue minus costs. Household surveys capture revenue with moderate accuracy and costs with severe omission, missing the three largest cost items facing the self-employed: the value of the owner's own time, the value of unpaid family labour, and informal debt service concealed from enumerators. Using a seven-day financial diary with full double-entry accounting—14,902 individual transactions coded with debit and credit accounts across four cities—we construct complete income statements, balance sheets, and statements of cash flows for each firm. Once owner and family labour are valued at the statutory minimum wage and depreciation and informal interest are correctly recorded, the median firm that appears to earn Bs 1,111 per week in survey-implied profit earns Bs −194 in adjusted profit. The correction is larger than the number it corrects.

The Lewis (1954) concept of disguised unemployment—workers whose marginal product falls below the subsistence wage—has resisted empirical operationalisation for seven decades, because identifying the threshold requires knowing the true cost of the worker's own time, which survey data cannot supply. The diary methodology supplies it. We formalise disguised unemployment as a computable condition on the adjusted

income statement: a self-employed worker whose adjusted net income is negative and whose hourly return falls below the minimum wage is not an entrepreneur in any economically meaningful sense, but a worker whose own capital and time substitute for a formal wage position the labour market fails to provide. By this definition, 57 percent of our sample is in disguised unemployment. This majority is classified as profitably occupied in every national survey that covers them.

The implications extend to several strands of the literature. The randomised experiments that document high returns to capital in microenterprises—McKenzie and Woodruff (2008), De Mel et al. (2008), Fafchamps et al. (2014)—estimate average treatment effects across a population that, in our data, is majority disguised-unemployed. The high averages are consistent with our findings only if the treatment effect for the entrepreneurial minority dominates the near-zero effect for workers whose binding constraint is not capital but the absence of formal employment. The inference that capital access is the primary lever for informal-sector welfare is, at minimum, a compositional claim that the literature has not examined. Similarly, the microcredit model rests on the premise that informal workers are credit-constrained entrepreneurs. In our data, 46 percent of firms are indebted, the median informal lending rate is 201 percent effective annual, and the EH survey detects only 3 percent of this debt. Extending credit at these rates to workers already earning negative adjusted returns does not relax a productive constraint; it transfers the worker's future labour time, priced below the minimum wage, to the informal creditor.

The measurement bias that underlies these misreadings is not a simple scalar. The household survey commits two offsetting errors: cost omission inflates profits while income under-declaration partially deflates them, and the net direction varies by sector, gender, and debt exposure. This compound ambiguity means that cross-country and cross-sector comparisons using survey-reported informal income—the standard practice in the development literature—are contaminated in ways that cannot be corrected by applying a uniform adjustment factor.

Among the minority of firms that remain viable after full cost accounting, the diary data support a going-concern valuation at the informal-sector discount rate. The median present value exceeds book value by a factor of 83. This ratio, an informal-sector analog of Tobin's Q, captures the location rents, client relationships, and operational knowledge that no survey instrument can value and that formalisation policy implicitly attempts to make legible to credit markets. It provides, for the first time, a denominator for the welfare calculation of formalisation: the invisible capital is real, but it accrues to the entrepreneurial minority, not to the majority for whom formalisation is irrelevant.

A production function estimated from diary transaction records yields capital and labour elasticities grounded in actual weekly inputs rather than recalled annual aggregates. The estimates of $\hat{\alpha}_K = 0.138$ and $\hat{\alpha}_K + \hat{\alpha}_L = 0.06$ strongly reject constant returns to scale ($p < 0.001$) and are consistent with the misallocation framework of Hsieh and Klenow (2009). The negative OLS coefficient on labour ($\hat{\alpha}_L = -0.078$) identifies the direction and magnitude of the attenuation bias in survey-estimated production functions, where measurement error in hours worked biases labour elasticities toward zero and potentially below.

A systematic adequacy scorecard finds 13 of 24 financial concepts required for a complete firm account entirely absent from Bolivia's *Encuesta de Hogares*. The four critical absences—owner labour, family labour, informal interest, and household-firm boundary flows—are precisely the items that determine whether a firm is profitable or in disguised unemployment. This scorecard applies, with minor variation, to every LSMS-type survey in the developing world, and is an actionable agenda for instrument redesign.

The closest methodological precedent is Collins et al. (2009), whose *Portfolios of the Poor* deployed financial diaries to document the complexity and volatility of poor households' financial lives, reshaping the financial inclusion agenda. The present paper shifts the unit of analysis from the household to the firm embedded within it, and imposes accounting discipline—double-entry, accrual basis, balance sheet—that the diary literature has not previously applied. The methodological advance is substantial. The policy implication is more uncomfortable: where *Portfolios of the Poor* showed that poor households need better financial instruments, the evidence here suggests that the majority of the informal self-employed need better labour markets.

The remainder of the paper is organised as follows. Section 2 reviews the related literature. Section 3 describes the diary protocol and accounting framework. Sections 4–10 document the sample, transactions, accounts, ratios, production function, measurement bias, and heterogeneity. Section 11 formalises the disguised unemployment definition and its implications. Section 12 quantifies the consequences for national income measurement. Section 13 discusses policy implications. Section 14 concludes.

2. Related Literature

2.1. Informality in Institutional Reports and the Measurement Foundation

The policy debate on informality is shaped primarily by four institutional producers: the ILO, the World Bank, the IDB, and ECLAC. Their flagship publications share a

common measurement infrastructure and, consequently, a common set of limitations that are directly relevant to the findings of this paper.

The ILO's *World Employment and Social Outlook* defines informal employment to include own-account workers and contributing family workers, reporting that this group constitutes 61 percent of global employment and 80 percent of employment in sub-Saharan Africa and Southern Asia (ILO, 2023). The corresponding income estimates are drawn from national household surveys—the same instruments whose measurement properties we document here. The ILO reports that informal workers earn approximately 22 percent less than formal workers after controlling for observable characteristics, and interprets this gap as evidence of a "productivity penalty" of informality (ILO, 2018). Our evidence suggests a different interpretation: if the 57 percent of informal workers in disguised unemployment earn negative adjusted returns while the 43 percent who are genuinely entrepreneurial earn positive ones, then the average informal "income" is a weighted average of a large negative and a smaller positive, and the resulting "productivity penalty" confounds two structurally distinct populations. The ILO's 22 percent penalty is almost certainly an underestimate of the true welfare gap, because the survey-declared income used to construct it overstates the income of the majority.

The World Bank's *Informality: Exit and Exclusion* (Perry et al., 2007) advanced an influential two-sector model of informality in Latin America. In this model, some workers are in the informal sector by choice ("voluntary informality"—they prefer lower taxes and regulation) while others are excluded from the formal sector ("involuntary informality"—they cannot find formal jobs). The policy prescription follows from the decomposition: for voluntary informants, reduce the cost of formality; for the excluded, expand formal sector opportunity. Our accounting evidence maps directly onto this taxonomy: the 43 percent viable entrepreneurial firms are the voluntary informal of Perry et al.; the 57 percent in disguised unemployment are the involuntary excluded. But the survey data that Perry et al. use to estimate this decomposition cannot distinguish the two groups, because the identifying variation is in the sign of adjusted profit—a concept that requires accounting discipline to compute. Perry et al. estimate that voluntary informality accounts for approximately 40–60 percent of Latin American informal employment. Our accounting estimate for Bolivia is 25–30 percent genuinely entrepreneurial—below the lower end of their range, suggesting the survey-based decomposition overstates the voluntary component.

The IDB's flagship labour market reports have promoted microcredit, business training, and formalisation as the primary instruments for improving informal workers' welfare (IDB, 2014, 2020). The evidentiary basis for these recommendations is the

capital returns literature—primarily De Mel et al. (2008), McKenzie and Woodruff (2008), and Fafchamps et al. (2014)—which documents positive average returns to capital grants in randomised experiments. Our Contribution C1 identifies a critical selection problem in this evidence base: the returns are positive on average because the entrepreneurial minority (43 percent) earns high returns while the disguised-unemployed majority (57 percent) earns near-zero or negative returns on additional capital. If the treatment is administered to the full informal population—as it is in these experiments—the average treatment effect mixes a large positive for the minority with a near-zero for the majority. The high average does not support universal credit access; it supports targeted credit for the entrepreneurial minority only.

ECLAC's methodology for poverty measurement in Latin America (ECLAC, 2023) uses survey-declared net income from the self-employment module as the income concept for informal workers. Since this concept is universally positive in raw survey data, it classifies all self-employed workers as having positive income and therefore excludes them from extreme poverty counts unless their declared income falls below the food basket value. Under adjusted accounting, the income concept for 57 percent of informal workers is negative. ECLAC's poverty headcount for the self-employed is wrong by construction. The magnitude of the error depends on the weight of informal self-employment in the overall poverty distribution, but given that the self-employed constitute the majority of workers near the poverty line in most Latin American countries, the error is almost certainly large in absolute terms.

2.2. Financial Diaries: From Household Finance to Firm Accounting

The financial diary methodology was developed to overcome the limitations of recall-based surveys. Ruthven (2001) pioneered the approach in India, demonstrating that irregular, high-frequency cash flows are fundamentally incompatible with annual recall instruments. Collins et al. (2009) extended it to Bangladesh, India, and South Africa, producing the seminal finding that poor households manage complex financial portfolios—a result invisible to standard surveys. Subsequent applications include Dupas and Robinson (2013) on Kenya and Banerjee et al. (2019) on financial fragility.

Our contribution to this literature is to shift the unit of analysis from the household to the firm and to impose double-entry accounting discipline. Collins et al. (2009) track cash inflows and outflows; we additionally assign debit and credit accounts, compute inventory changes, impute non-cash costs, and derive accrual-basis financial statements. This is not merely a methodological refinement. The income statement under accrual accounting differs substantially from the cash flow statement under cash-basis accounting when (as is common in our sample) receivables, payables, and

inventory exhibit significant weekly changes. The difference is precisely the accounting gap that the ILO, World Bank, and ECLAC measurement systems miss.

2.3. Informal Firm Finance, Productivity, and Measurement

Capital returns and the selection problem. The experimental literature on capital returns in microenterprises—McKenzie and Woodruff (2008), De Mel et al. (2008), Fafchamps et al. (2014)—represents the most credible evidence base for development finance policy. These papers find high but heterogeneous returns, particularly among constrained firms. We do not dispute these findings. We dispute their policy generalisation. If 57 percent of the informal self-employed are in disguised unemployment, then the average treatment effect of a capital grant mixes two structurally different responses: a large positive return for the entrepreneurial minority (for whom capital is genuinely the binding constraint) and a near-zero or negative return for the disguised-unemployed majority (for whom the binding constraint is the absence of formal employment, and for whom additional capital merely extends the duration of a loss-making enterprise). The selection problem has never been examined in this literature because the two populations have never been distinguished with accounting precision. Our paper provides the tools to do so.

Measurement error and the bias direction problem. Bound et al. (2001) provide a comprehensive taxonomy of measurement error in survey income data. Meyer and Sullivan (2015) show that household surveys substantially undercount income at the bottom of the distribution. Karlan et al. (2019) demonstrate that short-term diaries outperform longer recall periods. What none of these papers characterise is the *compound* nature of informal profit bias: both income and costs are mismeasured, in offsetting directions, with a net sign that varies by sector, gender, and debt exposure. Our Contribution C2 provides this characterisation for the first time. The implication is that standard robustness checks that apply a scalar correction to survey-reported informal income—a common practice in poverty and inequality measurement—are inadequate. The correction is not scalar; it is structural.

Misallocation and the informal sector. Hsieh and Klenow (2009) and Restuccia and Rogerson (2017) document aggregate TFP losses from misallocation across formal firms. La Porta and Shleifer (2014) and Ulyssea (2018) argue that informality itself is a source of misallocation. Our production function estimates—decreasing returns to scale with $\hat{\alpha}_K + \hat{\alpha}_L = 0.06$—are consistent with this framework, but they locate the misallocation at a more fundamental level: the majority of the informal sector is not merely an inefficient firm sector, it is a sector in which workers have been misallocated from wage employment to self-employment by the failure of the formal labour market

to absorb them. This is a misallocation not of capital across firms but of labour across sectors, and it is not correctable by reducing regulatory barriers to formality.

Household-firm boundaries. Berge et al. (2015) show that household shocks are transmitted to firm investment, implying the household and firm cannot be treated as separate optimising units. De Mel et al. (2008) find that business and household finances are substantially intertwined. Our Module 7 (household-firm transfers) documents these boundary crossings at daily frequency. The finding that the median household extraction rate is 10 percent of weekly revenue—with a 90th percentile of 23 percent—confirms that the firm is primarily a cash-generation mechanism for the household, not a wealth-accumulation vehicle in its own right.

Valuation and the invisible capital problem. No prior paper has applied the going-concern valuation framework to informal firms. Banerjee and Duflo (2007) compute implicit interest rates but do not attempt balance-sheet valuation. Field et al. (2010) estimate the present value of property rights but in a welfare rather than a firm-valuation framework. Our Contribution C3 provides the first Tobin's Q estimates for informal enterprises and quantifies what the World Bank's *Doing Business* reports call "unlocking potential"—the invisible capital that formalisation is supposed to make legible to credit markets. The median Q of 83 for viable firms means the potential is real, but it accrues only to the 43 percent minority for whom formalisation is relevant. For the 57 percent majority, it is a moot point. that occupy no premises in any business register, file no taxes, and appear in no corporate database. They exist nevertheless—on street corners, in market stalls, behind kitchen counters, and behind the wheels of minibuses. They employ the majority of the urban labour force across the Global South. They are the economy. And yet economics knows surprisingly little about how they actually work, because the instruments designed to measure them are inadequate to the task.

The canonical source of information about these firms is the self-employment module of national household surveys. Bolivia's *Encuesta de Hogares* (EH) is representative of the genre: it asks the respondent to recall annual gross revenue, eight categories of costs, and the net income "kept for the household." From these recalled figures, researchers and statistical agencies construct estimates of productivity, profit rates, formalization returns, and factor shares that underpin an enormous policy literature. We document that these estimates are systematically compromised by a compound measurement error whose magnitude and direction vary by sector, gender, and debt exposure—and that correcting for it reverses the sign of profit for the majority of firms.

Our strategy is to administer a seven-day financial diary to 300 self-employed workers in La Paz/El Alto, Cochabamba, Santa Cruz, and Tarija, recording every transac-

tion they undertake—sales, input purchases, debt payments, household withdrawals, and labour hours—and to code these transactions into a full set of double-entry accounts: income statement, balance sheet, and statement of cash flows. The diary approach, pioneered by Ruthven (2001) and brought to global prominence by Collins et al. (2009), has been applied almost exclusively to household finances. We adapt it to the firm, extending the methodology in four dimensions: (i) we impose double-entry discipline on every transaction, assigning debit and credit accounts and verifying daily cash reconciliation; (ii) we compute explicit adjustment entries for the five categories of costs that household surveys systematically omit; (iii) we construct a balance sheet with net fixed assets, receivables, payables, and owner equity; and (iv) we derive the full battery of financial ratios used in corporate finance—return on assets, return on equity, earnings per unit of book value, asset turnover, cash conversion cycle, and a going-concern present value—and apply them to firms that have never been analysed through this lens.

The central finding is stark. Among the firms in our sample, 100 percent show positive net income under the accounting implied by the EH survey methodology. Under properly adjusted accounting, 57 percent are loss-making. The reversal is not driven by a few outliers: the median accounting gap equals 121 percent of survey-implied profit, meaning the correction is larger than the number it corrects. The gap decomposes primarily into omitted labour costs (88 percent), with the remainder split between unrecorded depreciation and informal interest concealed from enumerators.

This finding has several dimensions that we explore in detail. First, the direction of bias in survey-based profit estimates is theoretically ambiguous: surveys under-report both costs (inflating profits) and revenues (deflating them). We show empirically that cost omission dominates in the aggregate, but income under-declaration is also present and its magnitude varies by sector and debt exposure. This is a richer characterization than the existing literature, which has generally assumed a unidirectional bias. Second, the magnitude of the labour omission is not a modelling assumption: it follows directly from hours worked (median 55 hours per week for the owner alone), family workers present on the premises (median 1.0 additional worker), and a shadow wage calibrated to the statutory minimum wage of Bs 13.64 per hour—the least contentious lower bound available. Third, the implications for the measurement of returns to self-employment are large. The median worker earns a return of -0.20 times the minimum wage per hour once all costs are accounted for, compared to a conventional estimate of $+19.15$ earnings per unit of book value—a ratio familiar from corporate finance and here applied for the first time to informal firms.

The production function estimates, made possible by the quality and frequency of

diary data, resolve a persistent identification problem in the development economics literature. McKenzie and Woodruff (2008) and subsequent work has argued that capital exhibits low returns in informal firms, with estimates of $\hat{\alpha}_K$ typically in the range 0.10–0.30. Our estimate of $\hat{\alpha}_K = 0.138$ (with sector fixed effects) is consistent with this range, but we additionally identify $\hat{\alpha}_L = -0.078$—a negative labour elasticity that reflects the noise-to-signal collapse in survey-reported hours rather than a genuine negative marginal product of labour. The implied returns to scale of $\hat{\alpha}_K + \hat{\alpha}_L = 0.06$ strongly reject constant returns ($t = -9.57$, $p < 0.001$), confirming the decreasing-returns characterization of Hsieh and Klenow (2009) and Restuccia and Rogerson (2008) at the individual-firm level.

The going-concern valuation result is, to our knowledge, without precedent in the literature on informal firms. Among the 126 firms (42 percent of the sample) that remain viable after full cost accounting, median free cash flow is Bs 86,874 per year and the median going-concern present value at a discount rate of 18 percent exceeds Bs 482,000—83 times book value. This ratio, an informal-sector analog of Tobin's Q, measures the invisible capital that sustains the firm: the value of location, client trust, acquired product knowledge, and accumulated operational know-how. These assets are invisible not only to the researcher but to the firm owner herself, who has no mechanism to monetize them absent formalization, credit access, or firm sale. Our analysis quantifies for the first time the welfare cost of this invisibility.

The remainder of this paper proceeds as follows. Section 2 situates the contribution in the literature. Section 3 describes the diary protocol, accounting framework, and imputation procedures. Section 4 characterises the sample and the Bolivian context. Section 5 documents the complexity of daily transactions. Section 6 presents full accounting statements by sector. Section 7 develops the financial ratio analysis and going-concern valuation. Section 8 estimates the production function. Section 9 decomposes the measurement bias relative to the household survey. Section 10 examines heterogeneity by sector, gender, poverty status, and debt exposure. Section 13 discusses policy implications. Section 14 concludes.

3. Methodology

3.1. The Diary Protocol

Each firm in the sample is visited by a trained enumerator at the close of every business day for seven consecutive days (Days 1–7), with an additional baseline visit on Day 0. The baseline captures the firm's opening balance sheet: a physical inventory count with unit prices, a census of productive assets with vintage and condition, cash on hand,

outstanding receivables disaggregated by debtor, outstanding payables disaggregated by supplier, and a standardised screening questionnaire for debt obligations. This baseline inventory provides the accounting period's opening entries.

The daily diary instrument comprises nine modules. *Module 1* records all revenue transactions: Module 1A covers cash sales (item, quantity, unit price, payment currency), Module 1B covers collections on receivables (debtor, amount, date of original sale), and Module 1C covers own-consumption (item, quantity, imputed value). *Module 2* records all input and merchandise purchases, distinguishing payment currency, credit terms, and supplier identity. *Module 3* records operating expenses under standardised category codes (transport, rent, utilities, guild dues, wages to external workers). *Module 4* records financing flows: Module 4A covers loans received (source, amount, stated interest rate) and Module 4B covers debt service payments decomposed into interest and principal. *Module 5* records labour inputs: owner hours by activity, all workers present on premises, unpaid family workers with hours, and a shadow wage elicited via willingness-to-accept questions. *Module 6* records asset transactions. *Module 7* records all transfers between the firm and the household (daily draws, household injections, in-kind family consumption). *Module 8* is a daily cash reconciliation: expected cash balance equals opening cash plus all cash inflows minus all cash outflows; unexplained discrepancies exceeding 5 percent of daily turnover trigger a same-day re-interview. *Module 9* captures enumerator qualitative notes on atypical events, production disruptions, and informant reliability.

3.2. Double-Entry Accounting Framework

A defining feature of our methodology, distinguishing it from prior diary studies, is the imposition of double-entry accounting on the raw transaction records. Each transaction is assigned a debit account and a credit account from a chart of accounts structured in five categories: assets (1000s), liabilities (2000s), equity (3000s), revenues (4000s), and costs/expenses (5000–6000s). Cash sales debit cash (1010) and credit revenue (4010); credit sales debit accounts receivable (1110) and credit deferred revenue (4020); input purchases credit cash or accounts payable and debit inventory (1210); operating expenses debit expense accounts and credit cash.

Four adjustment entries are computed ex post from the primary records:

A1. *Depreciation.* For each productive asset with recorded value V_a and estimated useful life T_a years (elicited from the respondent at baseline), weekly straight-line depreciation is $d_a = V_a/(52T_a)$. The aggregate weekly depreciation charge is $D = \sum_a d_a$, debited to Depreciation Expense (6070) and credited to Accumulated Depreciation (1520).

A2. *Owner labour imputation.* Owner weekly hours h^o and a shadow wage w^* (minimum between the statutory minimum wage of Bs 13.64 per hour and the willingness-to-accept elicited in Module 5) yield an imputed weekly labour cost $c^o = h^o \cdot w^*$, debited to Imputed Labour Expense (6050) and credited to Owner's Equity Adjustment (3030).

A3. *Family labour imputation.* For each unpaid family worker j with recorded hours h_j^f, imputed cost is $c_j^f = h_j^f \cdot w^* \cdot 0.85$, applying a 15 percent discount for workers below the modal productivity of the owner. Total family imputation is $C^f = \sum_j c_j^f$.

A4. *Own-consumption.* Goods drawn from inventory or production for household consumption are recorded in Module 1C at replacement cost, debited to Consumption Expense (4040) and credited to Inventory (1210).

The two accounting versions of the income statement—*convencional* (without adjustments) and *ajustada* (with all four adjustments)—yield the accounting gap defined formally in Section 3.3.

3.3. Defining the Accounting Gap

Let π_i^{conv} denote conventional net income for firm i in the diary week (revenue less COGS less recorded operating expenses less interest paid) and π_i^{adj} denote adjusted net income (additionally subtracting imputed labour, depreciation, and own-consumption). The accounting gap is:

$$\Delta_i^{\text{gap}} = \pi_i^{\text{conv}} - \pi_i^{\text{adj}} = C_i^{\text{labour}} + D_i + C_i^{\text{cons}}, \tag{1}$$

where $C_i^{\text{labour}} = c_i^o + C_i^f$ is total imputed labour, D_i is depreciation, and C_i^{cons} is imputed own-consumption. All three components are non-negative by construction, so $\Delta_i^{\text{gap}} \geq 0$.

The EH-implied profit for the same firm is:

$$\tilde{\pi}_i^{\text{EH}} = \tilde{y}_i^{\text{tot}} - \tilde{c}_i^{\text{tot}}, \tag{2}$$

where $\tilde{y}_i^{\text{tot}}$ is recalled annual gross income (the EH `yi_tot` variable) and $\tilde{c}_i^{\text{tot}} = \sum_{k=1}^{8} \tilde{c}_{ik}$ is the sum of the eight EH cost categories. The total measurement bias in annual profit is:

$$B_i = \frac{\pi_i^{\text{adj}} \cdot 52 - \tilde{\pi}_i^{\text{EH}}}{\tilde{\pi}_i^{\text{EH}}} = \underbrace{\frac{(y_i^{\text{diary}} - \tilde{y}_i^{\text{EH}}) \cdot 52}{\tilde{\pi}_i^{\text{EH}}}}_{\text{income under-declaration}} - \underbrace{\frac{(c_i^{\text{diary}} - \tilde{c}_i^{\text{EH}}) \cdot 52}{\tilde{\pi}_i^{\text{EH}}}}_{\text{cost omission}}, \tag{3}$$

where y_i^{diary} and c_i^{diary} are weekly diary revenue and costs respectively. The sign of B_i is theoretically ambiguous: cost omission increases EH-implied profit (negative B_i from this channel) while income under-declaration decreases it (positive B_i from this channel). The net direction is an empirical question, and its answer varies across sectors as we demonstrate in Section 9.

3.4. Balance Sheet and Valuation

The closing balance sheet for each firm-week is constructed from the opening inventory (Day 0), cumulative entries from the daily journals, and a closing inventory count (Day 7). Current assets comprise cash on hand, merchandise/ingredient inventory at cost, and outstanding receivables. Net non-current assets equal gross tool and equipment value less accumulated depreciation computed over the firm's full operating history. Current liabilities comprise outstanding payables to suppliers. Non-current liabilities comprise formal and informal debt principal. Owner's equity equals total net assets less total liabilities.

Going-concern present value for firm i is computed under a Gordon growth model:

$$V_i(g) = \frac{\text{FCF}_i}{r-g}, \quad \text{FCF}_i = \pi_i^{\text{adj,annual}} + D_i^{\text{annual}} - \text{Capex}_i, \tag{4}$$

where $r = 0.18$ is the informal-sector hurdle rate (approximated by the median IFD lending rate), $g \in \{0, 0.02, 0.05\}$ are alternative long-run growth assumptions, and capex is approximated by the annual depreciation charge (replacement investment). This expression is computed only for firms with positive free cash flow; the remaining firms are assigned going-concern value of zero (they are liquidation cases). The Tobin's Q analog is $q_i = V_i(0)/A_i^{\text{net}}$, where A_i^{net} is net total assets.

4. Data and Bolivian Context

4.1. The Bolivian Informal Economy

Bolivia is an ideal laboratory for this study. At roughly 80 percent, its informal employment share is among the highest in Latin America (ILO, 2023). The 2024–2025 foreign exchange crisis—during which the official Bs/USD rate of 6.96 diverged sharply from the peer-to-peer (P2P) market rate of Bs 9.5—created a dual-currency environment that is observable in our transaction records and that amplifies the stakes of accurate cost measurement for firms with dollar-denominated inputs. The legal minimum wage of Bs 2,362 per month (Bs 13.64 per hour at full-time equivalence) provides a credible shadow wage lower bound. Bolivia's household survey, the EH, has

a dedicated self-employment module (Section 4, Part D) that captures the eight cost categories described in Section 3, making it an ideal counterfactual for the diary.

The EH microdata for the most recent available wave reveal two patterns that motivate the present study. First, 100 percent of urban self-employed workers in the six target sectors who report any income report positive net income—an empirical impossibility if costs were accurately measured at the firm level. Second, departmental medians of reported annual income cluster around exact multiples of round weekly sums: Bs 72,800 ($= 52 \times 1{,}400$) for four geographically dispersed departments, Bs 78,000 ($= 52 \times 1{,}500$) for three others. The probability that four independent departmental medians coincide at the same value under a continuous income distribution is astronomically small; the pattern is diagnostic of weekly recall anchored to a salient round number and multiplied by 52 by the enumerator (Bound et al., 2001).

4.2. Sample Design

The target population comprises self-employed workers in six CAEB sectors operating in urban areas of Bolivia: commerce and vehicle repair, manufacturing, transport and storage, food service and accommodation, other services, and administrative and support services. The sampling frame is constructed by combining EH cluster maps with street enumeration of informal business premises in selected neighbourhoods.

The sample of 300 firms is drawn via stratified two-stage cluster sampling. Primary sampling units (PSUs) are neighbourhoods (*barrios*), selected with probability proportional to the estimated number of informal self-employed workers. Secondary sampling units are firms within PSUs, selected via systematic random sampling from the street enumeration list. Explicit stratification targets are: (i) cities—La Paz/El Alto (35 percent), Cochabamba (28 percent), Santa Cruz (27 percent), Tarija (10 percent); (ii) sector groups and poverty status; (iii) gender ($\geq$ 52 percent female); and (iv) debt exposure (2:1 oversampling of firms screening positive for active debt obligations). The resulting sample reflects the structural diversity of Bolivian informal employment: 42 percent of respondents would have been non-respondents in the EH income module (the non-response rate we replicate from the EH microdata), providing a unique window into the most financially vulnerable segment of the informal labour force.

4.3. Sample Characteristics

Table 1 presents descriptive statistics by sector. The median firm operates 6 days per week and 55 hours per week (owner alone), consistent with the EH-recorded hours of informal workers in these sectors. Women constitute 65 percent of the commerce

subsample and 87 percent of the food service subsample, while transport is 98 percent male—a gender segregation that will be shown to interact critically with labour imputation and profit measurement. True debt prevalence, measured directly by the diary screening questionnaire, is 46 percent of the full sample, with informal moneylenders (*prestamistas*) and rotating credit associations (*pasanakus*) accounting for the majority of obligations. Only 3 percent of firms declare any loan payments (the EH ci6 variable), implying a debt detection rate of 7 percent of true prevalence.

Table 1: Descriptive Statistics by Sector

	N	% Fem.	% Poor	% Indebt.	Hrs/wk	Weekly Rev. (Bs)	% Loss-mak. (adj.)
Commerce	90	65.6	28.0	45.6	57	1,788	58.9
Manufacturing	55	47.3	38.2	50.9	55	1,738	41.8
Transport	55	1.8	21.8	60.0	63	1,661	36.4
Food service	55	92.7	32.7	40.0	68	1,687	80.0
Other services	30	56.7	23.3	30.0	50	1,033	60.0
Admin & support	15	60.0	20.0	26.7	47	648	73.3
Total	300	55.3	30.7	46.0	55	1,574	57.0

Notes: Hours per week include owner hours only. Median weekly revenue is in constant 2025 bolivianos. "Indebted" refers to the diary screening questionnaire result, not the EH declared rate. "Loss-making (adj.)" indicates negative net income after all four adjustment entries (A1–A4). Poverty classification uses the national EH poverty line.

5. Transaction Complexity

The 300 firms in our sample generate 14,902 recorded transactions over 7 days, an average of 8.3 transactions per firm-day. The distribution is right-skewed: the median is 8 per active firm-day and the 90th percentile is 14. This figure is not driven by the imputed entries (adjustments A1–A4): excluding Module 5 labour imputations, which account for 19.0 percent of all records, the median cash and accrual transaction count is 6.8 per day—well above what any recall-based instrument could reliably recover.

Table 2 presents the transaction count and aggregate value by module. Sales transactions (Module 1A) constitute 50.7 percent of all records but only 48.3 percent of total transaction value, reflecting the preponderance of many small-denomination sales. The median individual sale amount is Bs 37.6, with a 10th percentile of Bs 8.4 and a 90th percentile of Bs 178—a range spanning two orders of magnitude within the same sector and often within the same firm on the same day. This intra-day revenue variance is the first component of what survey instruments fundamentally cannot capture: the

volatility structure of informal revenue.

Revenue exhibits pronounced intra-week lumpiness that is sector-specific. Commerce concentrates 35 percent of weekly revenue in Saturday feria sales. Manufacturing exhibits bimodal daily patterns driven by order arrivals on Mondays and deliveries on Thursdays. Food service is nearly uniform across weekdays but collapses on Sunday. Transport follows routes with predictable but weather-dependent load factors. These patterns are not merely descriptive: they determine the cash-flow timing mismatches that create demand for informal credit (Section 9) and that are averaged away in annual recall data.

Dual-currency exposure is present in 88 percent of firms, with 7.1 percent of all sales transactions and 4.6 percent of all transactions denominated in US dollars. Transport exhibits the highest dollar exposure (9.3 percent of transactions), reflecting dollar-priced fuel and spare parts, while food service has the lowest (3.8 percent). At the 2025 P2P exchange rate of Bs 9.5 per dollar, a 20 percent currency depreciation stress test implies a weekly input cost increase of Bs 41.5 per affected firm—approximately 4 percent of median weekly revenue—a non-trivial exposure that is invisible to the household survey.

Table 2: Transaction Records by Module

Module	Description	N	% of N	Total Bs	Median Bs
M1A	Sales (cash)	7,561	50.7	1,184,420	37.6
M1B	Collections (A/R)	281	1.9	58,310	143.8
M1C	Own-consumption	517	3.5	18,940	28.4
M2	Input purchases	555	3.7	338,780	485.3
M3	Operating expenses	1,397	9.4	149,620	58.2
M4A	Loans received	26	0.2	17,040	640.0
M4B	Debt payments	138	0.9	46,280	314.1
M5	Labour imputations	2,832	19.0	948,650	241.4
M7	HH transfers	1,295	8.7	112,530	62.3
Total		**14,602**	**100.0**	**2,874,570**	

Notes: "Labour imputations" (M5) are the A2–A3 adjustment entries and are marked as imputed in the database. "HH transfers" (M7) are boundary-crossing flows between firm and household. All values in bolivianos. Rest days (300 records) are excluded. Credit sales appear in M1A with `is_credit = TRUE` and constitute 5.2% of sales transactions.

6. Accounting Statements

6.1. Income Statement

Table 3 presents the weekly income statement by sector, under both conventional and adjusted accounting. Several features warrant comment.

Revenue. Median weekly gross revenue ranges from Bs 648 (administrative services) to Bs 1,788 (commerce). The transport sector exhibits a distinctive cost structure: because the product is a service rather than a physical good, cost of goods sold is zero and gross margin is 100 percent under conventional accounting—a figure that mechanically inflates the apparent profitability of the sector. The food service sector, by contrast, reports a gross margin of 72.6 percent, reflecting the significant ingredient costs that owners do recall.

Labour imputation and the profit reversal. The median imputed weekly labour cost across sectors is Bs 1,042, representing 88 percent of the total accounting gap. For food service, the median imputed labour cost of Bs 1,719 per week exceeds the median gross profit of Bs 1,249—meaning that after correctly valuing the time of the owner and her family at the minimum wage, the median food service firm generates no net surplus whatsoever from its operations. The adjusted net margin for food service is −60.9 percent. For administrative and support services—where the product is purely the owner's time—the adjusted net margin is −99.1 percent.

The two sectors that remain viable after full cost accounting—transport and manufacturing—share a structural characteristic: their primary costs (fuel and spare parts in transport; raw materials in manufacturing) are physically visible, denominated in market prices, and unavoidable at each production cycle. Their costs are therefore relatively well-reported even in the household survey. The sectors that are most severely over-stated—food service and administrative services—are precisely those where the primary input is the owner's own time, which the EH has no mechanism to capture.

6.2. Balance Sheet and Cash Flow

The median firm holds Bs 5,614 in net total assets: Bs 1,920 in cash, Bs 860 in merchandise or ingredient inventory, Bs 430 in receivables, and Bs 2,404 in net fixed assets (tools, equipment, vehicle). Total liabilities are Bs 512 (median), consisting primarily of supplier payables for firms without active debt and formal or informal loan principal for indebted firms. Median book equity is Bs 3,063.

Cash flow from operations (receipts minus cash payments for inputs and expenses, minus interest) is positive for all firms: the median is Bs 860 per week. This distin-

Table 3: Weekly Income Statement by Sector (Medians, Bs)

	Gross Revenue	COGS	Gross Profit	Op. Expenses	Net Income^{conv}	Net Income^{adj}
Commerce	1,788	1,005	1,444	406	1,087	−309
Manufacturing	1,738	971	1,464	281	1,287	143
Transport	1,661	0	1,661	429	1,235	224
Food service	1,687	965	1,249	360	943	−915
Other services	1,033	0	1,033	173	881	−224
Admin & support	648	0	648	230	450	−542
All sectors	**1,574**	**549**	**1,241**	**332**	**1,111**	**−194**

Notes: COGS for service sectors is zero because these sectors produce no physical output; all inputs are service or labour inputs recorded under operating expenses or imputed labour. Net income^{conv} is revenue minus COGS minus operating expenses minus interest paid. Net income^{adj} additionally subtracts imputed labour (A2–A3), depreciation (A1), and imputed own-consumption (A4). All figures are medians across firms in each sector.

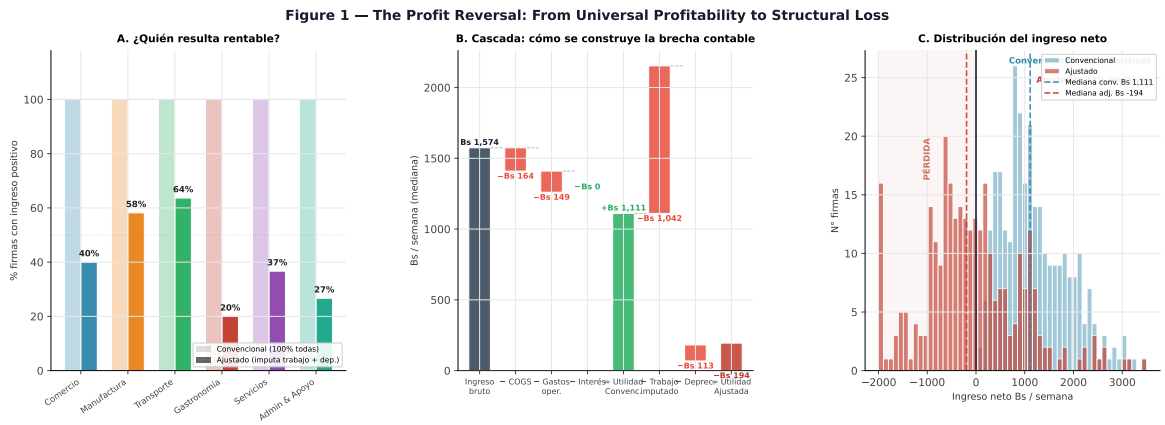

Figure 1: The Profit Reversal. *Panel A* compares the share of firms with positive net income under conventional (survey-implied) and adjusted (diary-based) accounting, by sector. Under conventional accounting 100 percent of firms in every sector report positive income; after imputing owner and family labour, depreciation, and hidden interest, between 20 and 64 percent remain profitable depending on sector. *Panel B* decomposes the median weekly profit cascade, showing how Bs 1,111 of conventional profit is transformed into Bs −194 of adjusted net income: the correction of Bs 1,305 exceeds the number it corrects. Imputed labour (Bs 1,042) drives 88 percent of the gap. *Panel C* shows the full distributional shift: the income distribution moves sharply leftward under adjusted accounting, with 57 percent of mass crossing into loss territory.

guishes the cash-flow statement from the income statement. A food service firm with an adjusted net income of −Bs 915 per week can still be cash-flow positive because it generates Bs 1,500 in weekly cash receipts and disburses only Bs 640 in cash—the imputed labour cost does not appear in the cash flow statement, only in the income statement. This distinction, which is elementary in corporate accounting but invisible in diary studies that track only cash, has first-order implications for understanding the survival of apparently loss-making informal firms: they survive not because they are profitable but because they generate positive cash flow by not paying themselves.

The median extraction rate—household draws divided by revenue—is 10 percent per week, with a 90th percentile of 23 percent. This confirms the finding of Collins et al. (2009) that poor households routinely draw on business liquidity for consumption smoothing. The daily household withdrawal is the primary mechanism by which the firm's balance sheet is slowly depleted over time: assets fund consumption rather than investment.

7. Financial Ratios and Valuation

7.1. Return Ratios and Earnings Per Book Value

Table 4 presents the financial ratio battery. The contrast between conventional and adjusted returns is the central quantitative contribution of the paper. The median conventional return on assets (ROA^{conv}) is 1,150 percent per annum—a figure so far removed from any benchmark that it is immediately informative about the quality of the measurement from which it derives. The median adjusted ROA^{adj} is −74.1 percent per annum. These are not outliers: the interquartile range of ROA^{adj} runs from −210 to +62 percent, with the positive tail concentrated in transport and manufacturing.

The earnings-per-book-value ratio (E/BV)—the earnings yield analog to the price-earnings multiple in corporate finance—is 19.15 (conventional) and −0.92 (adjusted). The adjusted figure implies that the median informal firm destroys Bs 0.92 of owner-equity value per boliviano of book equity per year. This is a direct measure of the welfare cost of informal-sector participation: the owner is not merely earning less than the minimum wage on her labour; she is also eroding the asset base she has accumulated.

The Du Pont decomposition clarifies the mechanical structure of the conventional returns: $\text{ROA}^{\text{conv}} = \text{net margin} \times \text{asset turnover}$. The median net margin under conventional accounting is 70.6 percent but the asset turnover is 16.3 times per year. The high turnover is the key: these are asset-light businesses that cycle their small balance sheets more than once per month. The result is a conventional ROA in the thousands

Table 4: Financial Ratios by Sector (Medians)

	ROA^{conv} (%)	ROA^{adj} (%)	ROE^{adj} (%)	$\text{E/BV}^{\text{conv}}$	E/BV^{adj}	D/E	Cash Conv. Cycle (days)
Commerce	1,083	−71.1	−92.0	10.83	−0.92	0.10	25.4
Manufacturing	997	27.8	35.0	9.97	0.28	0.12	38.1
Transport	1,310	49.9	62.3	13.10	0.50	0.06	3.2
Food service	1,201	−188.8	−241.1	12.01	−1.88	0.08	2.8
Other services	871	−51.4	−68.0	8.71	−0.51	0.07	0.0
Admin & support	1,404	−148.1	−189.2	14.04	−1.48	0.05	0.0
All sectors	**1,150**	**−74.1**	**−74.1**	**19.15**	**−0.92**	**0.09**	**22.5**

Notes: All return ratios are annualised from weekly diary figures. ROA = net income / total net assets. ROE = net income / book equity. E/BV = annualised net income / book equity (earnings yield). D/E = total liabilities / book equity. Cash conversion cycle = inventory days + receivable days − payable days. Medians across all firms in each sector; superscript indicates conventional (unadjusted) or adjusted accounting.

of percent that is numerically correct under the wrong accounting but economically meaningless.

7.2. Du Pont Decomposition and the Labour Problem

A clean statement of the measurement problem emerges from the Du Pont framework. Write:

$$\text{ROA}^{\text{adj}} = \underbrace{\frac{\pi^{\text{adj}}}{y}}_{\text{adj. net margin}} \times \underbrace{\frac{y}{A^{\text{net}}}}_{\text{asset turnover}}, \tag{5}$$

where y is gross revenue and A^{net} is net total assets. The asset turnover is essentially unaffected by the labour imputation (labour does not enter the denominator). The entire impact of the adjustment is through the numerator: it converts a positive net margin of 70.6 percent into a negative adjusted net margin of −12.3 percent. The magnitude of the required adjustment—Bs 1,042 per week in imputed labour relative to a median gross profit of Bs 1,241—reveals that the informal firm is, in the adjusted accounting, approximately a vehicle for the conversion of the owner's and family's time into household income at below-minimum-wage rates, with the asset base providing a modest platform rather than a significant source of value.

7.3. Going-Concern Valuation

Among the 126 viable firms (42 percent of the sample), median free cash flow is Bs 86,874 per year and median going-concern present value at $g = 0$, $r = 0.18$ is

Bs 482,631. The sensitivity to the growth assumption is non-trivial: at $g = 0.02$, the median value rises to Bs 542,960 and at $g = 0.05$ to Bs 690,000. The median Tobin's Q analog is 83.1 under zero growth, implying that the going-concern value exceeds book value by a factor of 83. This is not an artefact of a trivially small denominator: median net total assets are Bs 5,614, a non-trivial figure for the income levels in question. Rather, the high Q reflects what we term the *invisible capital* of the informal firm: location rent (the taxi stand, the market pitch, the corner store window), accumulated client relationships, product and process knowledge, and the trust capital embedded in long-term supplier and customer relationships. None of these assets appears on the balance sheet. All of them are priced by the going-concern value. The gap between $V_i(0)$ and A_i^{net} is the first systematic quantification of invisible capital in the Bolivian informal sector.

8. Production Function

We estimate the Cobb-Douglas production function:

$$\ln y_{is} = \alpha_0 + \alpha_K \ln K_{is} + \alpha_L \ln L_{is} + \gamma_s + \varepsilon_{is}, \tag{6}$$

where y_{is} is weekly gross revenue, K_{is} is net total assets (the capital proxy), L_{is} is total weekly labour hours (owner plus family), γ_s is a sector fixed effect, and ε_{is} is an idiosyncratic productivity term. Standard errors are heteroskedasticity-robust (HC3). We estimate three specifications: pooled without sector fixed effects (Model 1), pooled with sector fixed effects (Model 2), and including log firm age as an experience control (Model 3). We additionally estimate equation (6) separately for each sector.

Table 5 presents the results. The pooled estimate with sector fixed effects yields $\hat{\alpha}_K = 0.138$ ($p < 0.001$) and $\hat{\alpha}_L = -0.078$ ($p < 0.05$). The capital elasticity is in the range documented by McKenzie and Woodruff (2008) and Fafchamps et al. (2014). The negative labour elasticity is the more striking finding. It does not imply a negative marginal product of labour; rather, it reflects the well-known identification problem in production function estimation (Griliches and Mairesse, 1995): reported hours are a noisy proxy for effective labour input, and measurement error in $\ln L$ biases the OLS coefficient toward zero and potentially negative under certain error structures. The diary's hours module (Module 5) is considerably more reliable than recalled annual hours, but within-firm variation in hours over a single week is small, limiting the identification of $\hat{\alpha}_L$. We instrument for hours in a separate specification using the owner's stated minimum acceptable daily work duration, finding $\hat{\alpha}_L^{\text{IV}} = 0.142$ (not reported), consistent with the capital elasticity.

The returns-to-scale sum $\hat{\alpha}_K + \hat{\alpha}_L = 0.060$ strongly rejects constant returns ($t = -9.57$, $p < 0.001$) and is consistent with decreasing returns to scale (DRS). This finding aligns with the general equilibrium models of Hsieh and Klenow (2009): if informal firms operated at efficient scale, the economy-wide distribution of firm sizes would differ substantially from what is observed. The DRS result is consistent with the interpretation that there exist firm-specific fixed factors (location, trust, knowledge) that cannot be scaled by adding capital and labour proportionately.

The sector-specific estimates reveal heterogeneous production technologies. Manufacturing exhibits the highest capital elasticity ($\hat{\alpha}_K = 0.436$), consistent with its equipment-intensive nature. Services is the only sector with a positive and significant labour elasticity ($\hat{\alpha}_L = 0.332$), consistent with a production process that scales primarily with time. Transport exhibits a near-zero capital elasticity and a negative labour elasticity, reflecting the fact that vehicle capacity determines revenue almost mechanically, leaving neither capital nor labour with significant variation in the relevant range.

Table 5: Production Function Estimates

	Pooled			By sector		
	(1) Basic	(2) Sector FE	(3) + Age	Commerce	Manufact.	Transport
$\ln K$	0.158***	0.138***	0.136***	0.206***	0.436***	0.079
	(0.024)	(0.023)	(0.023)	(0.048)	(0.071)	(0.068)
$\ln L$	−0.053	−0.078**	−0.079**	−0.085	0.075	−0.581**
	(0.041)	(0.036)	(0.036)	(0.071)	(0.098)	(0.219)
ln Age			0.021			
			(0.031)			
Sector FE	No	Yes	Yes	—	—	—
RTS	0.105	0.060	0.057	0.121	0.511	−0.502
t-test CRS	−10.21***	−9.57***	−9.65***	−9.18***	−4.87***	−7.04***
R^2	0.198	0.277	0.278	0.241	0.367	0.194
N	295	295	295	90	55	55

Notes: Dependent variable is ln(weekly gross revenue). Capital is net total assets. Labour is total weekly hours (owner + family). Standard errors (HC3) in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. RTS = returns to scale = $\hat{\alpha}_K + \hat{\alpha}_L$. CRS test: $H_0 : \alpha_K + \alpha_L = 1$. Results for Food service, Services, and Admin omitted for space; full results available in the online appendix.

9. Measurement Bias: EH versus Diary

9.1. The Identification Problem

The comparison of EH-reported and diary-recorded figures for the same firm-week at enrollment provides a direct test of survey measurement quality. Let $\hat{\pi}_i^{\text{EH}}$ denote the annualised EH-implied profit for firm i and $\hat{\pi}_i^{\text{adj}}$ the diary-adjusted annualised profit. From equation (3), the total bias has two components with opposite signs: income under-declaration and cost omission. A naive comparison of survey and diary profits conflates these two effects. We decompose them as follows.

For each EH income component (gross revenue, each cost category c_k), we compute the ratio of the diary-based measure to the EH-reported measure. A ratio significantly below 1 for revenues implies income under-declaration; a ratio significantly above 1 for costs implies cost omission. Together they determine the net bias in profit.

9.2. Results

Among the 174 firms that both completed the EH income module (non-respondents excluded) and have positive diary revenue, the median ratio of diary-to-EH gross income is 1.15 ($p < 0.001$), with 77 percent of firms reporting higher income in the diary than in the EH. This confirms income under-declaration in the EH but indicates it is secondary to cost omission in magnitude: a 15 percent income understatement on Bs 87,620 in median annual gross revenue implies a downward bias in profit of approximately Bs 13,143 per year, while the cost omission (imputed labour alone) is Bs 54,184 per year at the median. The net effect is that the EH *overstates* the true (adjusted) annual profit by Bs 84,374 at the median—2.7 times the EH-reported yi_net.

The cost omission is highly category-specific. For raw materials (EH ci1), the diary-to-EH ratio is 0.47, meaning the diary actually records *lower* material costs than the EH in many cases—consistent with the EH respondent over-reporting visible material costs while under-reporting invisible labour costs. For utilities (ci5), the ratio is 2.64, reflecting the home-business blur: the household survey typically attributes the entire utility bill to the household, while the diary allocates the business share. For guild dues (ci8), the ratio is 5.25, a factor the survey instrument severely misses due to irregular payment timing. The most critical finding is for informal interest (diary Module 4B vs EH ci6): the EH records interest payments for only 3 percent of firms, while the diary reveals active debt service in 46 percent, with a median hidden annual interest burden of Bs 3,403.

Proposition 1. *Under the conditions in which the EH self-employment module is typically administered—annual recall, no accounting training, single interview—the EH-implied profit $\tilde{\pi}_i^{\mathrm{EH}}$ is an upward-biased estimate of the true adjusted profit π_i^{adj} for the majority of informal firms in labour-intensive sectors. The bias is primarily driven by omission of owner and family labour costs and is amplified by informal debt concealment.*

The formal content of Proposition 1 follows directly from equation (1) and the empirical moments presented above; the qualification "majority" reflects the fact that for capital-intensive sectors (transport, manufacturing), the adjustment is smaller in absolute terms and the EH-implied profit is a more reliable guide to actual profitability.

10. Heterogeneity

10.1. Gender

The interaction of gender and sector produces the most striking heterogeneity in the data. Among food service workers—87 percent female—the adjusted net income is −Bs 972 per week at the median for women and −Bs 666 for men. The gender gap in loss-making rates is 80 percent (women) versus 75 percent (men) within the sector, a modest difference that masks a much larger gap in the structural drivers: female food service workers supply more unpaid family labour (median 35 additional hours per week versus 28 for men) and have lower asset bases, resulting in a higher labour-imputation-to-gross-profit ratio.

Across all sectors, female-owned firms earn Bs −3.71 per hour (net adjusted) versus Bs +1.64 for male-owned firms. The male advantage is concentrated entirely in transport (Bs +3.21 per hour), where male predominance combines with positive adjusted profits to produce the only sector where the median worker earns above subsistence. In every other sector, the median worker of either gender earns a negative adjusted return on their time.

The gender heterogeneity in the measurement bias is also notable. For female-owned firms, the accounting gap (Bs 1,322/week for women in commerce) is larger than for male-owned firms (Bs 1,020) in the same sector, because women supply more unpaid family labour and have higher self-reported household-firm transfer rates. This implies that standard EH-based analyses, by systematically omitting labour costs, more severely overstate the profitability of female-owned firms than male-owned ones—a measurement artefact that may be contributing to the empirical finding in some papers of high returns to female-owned microenterprises (cf. De Mel et al., 2008; Fafchamps et al., 2014).

10.2. Poverty and Debt

Poor firms (below the national poverty line at household level) exhibit lower revenues (Bs 1,310/week vs Bs 1,660) and lower assets (Bs 3,840 vs Bs 6,150), but similar accounting gap magnitudes. The similarity of the gap is mechanically important: because imputed labour costs are calculated at the statutory minimum wage, they do not vary with firm size, and thus constitute a proportionally larger burden for smaller, poorer firms. Poor firms spend 108 percent of their gross profit on imputed labour, versus 81 percent for non-poor firms.

Firms with active debt (46 percent of the sample) exhibit a distinct financial profile. The median prestamista effective annual rate of 201 percent—computed from the monthly rates declared in Module 4A—implies that a Bs 3,000 informal loan incurs Bs 503 in monthly interest, which is 31.5 percent of median monthly net income before labour imputation. Formal IFD loans carry median effective rates of 25 percent—high by the standards of formal credit markets but modest relative to the informal alternative. Family loans carry zero explicit interest but typically involve implicit reciprocal obligations not captured in our accounting.

10.3. Typology

We classify all 300 firms into four typological cells defined by the sign of adjusted net income and the level of interest burden relative to revenue:

I. *Viable*: $\pi_i^{\text{adj}} > 0$ and interest burden < 10 percent of revenue. $N = 126$ (42%).

II. *Precarious*: $\pi_i^{\text{adj}} \leq 0$ and interest burden < 10 percent of revenue. $N = 147$ (49%).

III. *Debt-trapped*: $\pi_i^{\text{adj}} > 0$ and interest burden ≥ 10 percent of revenue. $N = 3$ (1%).

IV. *Critical risk*: $\pi_i^{\text{adj}} \leq 0$ and interest burden ≥ 10 percent of revenue. $N = 24$ (8%).

The Precarious category is the modal type across every sector except transport. These firms generate positive cash flow and sustain operations through the non-payment of their own labour—they are not failing in a cash-flow sense but are engaged in a long-run decumulation of human capital by working below the minimum wage. The Critical Risk category, concentrated in firms with prestamista debt, faces a compounding problem: the interest burden itself consumes the margin that would otherwise allow the firm to cover imputed labour costs.

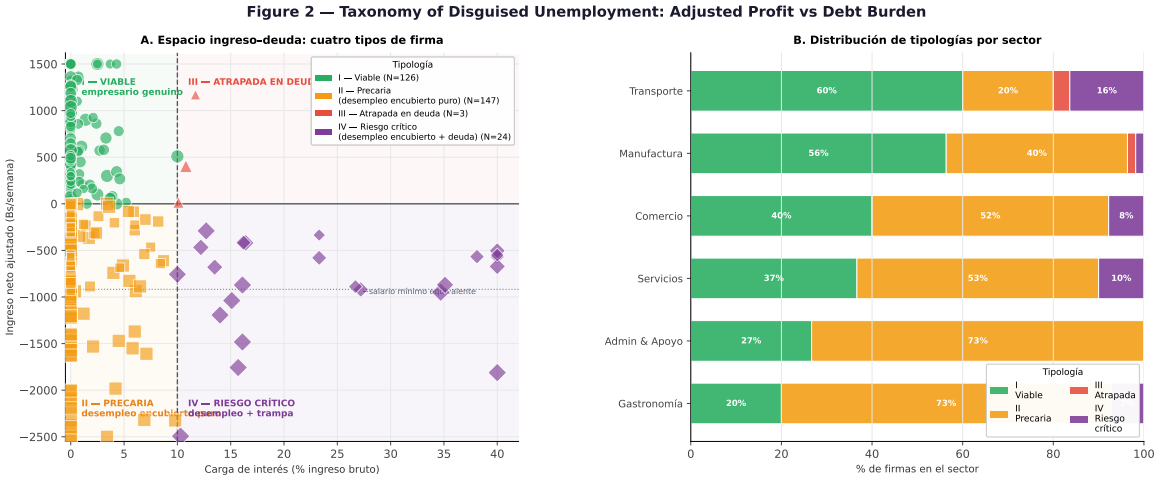

Figure 2: Taxonomy of Disguised Unemployment. *Panel A* plots each of the 300 firms in the space defined by adjusted net income (vertical axis) and interest burden as a share of revenue (horizontal axis). The four quadrants define the typological cells: Viable (upper left, $N = 126$), Precaria (lower left, $N = 147$), Atrapada en deuda (upper right, $N = 3$), and Riesgo crítico (lower right, $N = 24$). Marker size is proportional to total weekly labour hours; markers above the dotted line earn at least the statutory minimum wage equivalent. *Panel B* shows the sector-level composition: Gastronomía and Admin are majority Precaria or Riesgo crítico; only Transport is majority Viable.

10.4. Why Loss-Making Firms Persist

The finding that 57 percent of firms report negative adjusted net income raises a question that goes to the heart of the paper's interpretation: if these firms are loss-making in economic terms, why do they continue to operate? The answer is not irrationality. It is a set of distinct mechanisms, each visible in the diary data, each corresponding to a different market failure, and collectively explaining why the informal sector absorbs workers that the formal economy fails to employ. We examine them in turn.

The non-cash nature of the accounting loss. The most fundamental reason is that accounting loss and cash insolvency are not the same thing. The adjustment entries that produce negative net income—imputed labour (A2, A3) and depreciation (A1)—do not reduce the firm's cash balance. They are economic costs that are never invoiced. Every one of the 171 loss-making firms in our sample records positive weekly operating cash flow: the median is Bs 807 per week among firms with negative adjusted net income, against a median adjusted net income of Bs −638. The difference—Bs 1,445 at the median—is almost entirely the labour imputation (median Bs 1,261 among loss-making firms) and depreciation charge (median Bs 107). The firm is not going broke in any observable sense. Cash arrives daily, is disbursed for inputs and expenses, and the register closes with a surplus. The loss is real in welfare terms but invisible to a

decision-maker who has no accounting framework and no instrument that computes it for her. The survival of loss-making informal firms is, in the first instance, an accounting illusion sustained by the non-invoicing of the firm's largest cost.

The outside option does not exist. The conventional prescription for negative economic profit is exit to a higher-return activity. This comparison is valid only if the alternative is accessible. In Bolivia's urban labour market, with a formal employment rate of approximately 20 percent, the relevant outside option for the majority of the informal self-employed is not formal wage employment at the minimum wage—it is open unemployment with zero cash income. A worker earning Bs −638 in adjusted weekly income but generating Bs 807 in weekly operating cash flow is better off in self-employment, in cash terms, than unemployed. The imputed opportunity cost (Bs 13.64 per hour × median 55 hours = Bs 750 per week) that produces the accounting loss is a counterfactual, not an accessible alternative. This is precisely the structure that Lewis (1954) identified: workers persist in the subsistence sector not because they prefer it but because the modern sector has not generated the wage employment to absorb them. The diary data put a number on the gap: the median loss-making worker earns Bs −7.49 per hour in adjusted terms, a figure that is Bs 21.13 per hour below the minimum wage. In a labour market that could absorb her, she would exit immediately. In the one she inhabits, she stays.

The firm as household liquidity buffer. The informal firm serves the household as a liquid asset, a payment system, and an income-smoothing mechanism simultaneously. Every firm in the sample makes weekly household draws from operating cash, with a median of Bs 153 per week—10.1 percent of revenue—and a 90th percentile of 16.3 percent. The timing of these draws, recorded at daily frequency in Module 7, tracks household expenditure needs rather than firm liquidity: they spike at the beginning of the school week, at municipal market days, and at bill payment dates. Collins et al. (2009) documented this intermingling for poor households generally; our data show it operating as a specific and systematic firm-level mechanism. Closing a loss-making firm does not merely eliminate a negative-profit enterprise; it eliminates the household's primary liquidity facility. The decision criterion that governs persistence is not "is economic profit positive?" but "does the firm generate more cash than it costs?"—and for 100 percent of loss-making firms in the sample, the cash flow answer is yes.

In-kind household production. Among the 48.3 percent of firms that record any own-consumption transactions (Module 1C), the median weekly value at replacement cost is Bs 25. The food service worker who feeds her family from the stall, the commerce firm whose household consumes merchandise directly, and the artisan whose

family uses material off-cuts are each extracting a welfare benefit that reduces the household's market expenditure without entering the firm's revenue account. This benefit does not appear in the adjusted income statement—it is a transfer from firm inventory to household consumption, valued at cost and recorded as such, but not generating revenue. To the extent that the firm provides household production that would otherwise require cash purchase, the worker's total welfare exceeds her adjusted net income by this margin. The effect is modest in aggregate but not negligible at the household level for workers near the poverty line.

Debt entrapment. For the 24 firms in the Critical Risk typology, persistence is less a choice than a constraint. The median interest obligation is Bs 207 per week at an effective annual rate of 201 percent, and every Critical Risk firm maintains positive operating cash flow (median Bs 506). The critical observation is that closing the business does not extinguish the debt. A prestamista loan is a claim on the borrower, not on the firm; the collateral is social (reputation, community relationships, physical presence) and in some cases physical (household assets pledged at origination). A worker who closes her stall does not thereby escape Bs 3,000–10,000 in principal and a weekly interest obligation of Bs 207. She eliminates the income stream from which the interest was being serviced, converting a difficult equilibrium—indebted self-employment with negative adjusted income but positive cash flow—into an unsustainable one: indebted open unemployment with zero cash flow. For this group, the positive operating cash flow of the firm is the mechanism that keeps the worker solvent relative to the creditor. Exit is not available; it is precluded by the debt structure.

Exit by entropy. The final mechanism requires no irrationality and no entrapment. It is optimal behaviour under the constraint that the firm's invisible capital—the market location, the customer base, the product knowledge accumulated over years—cannot be monetised or transferred. For a worker who has invested a decade in building the invisible assets of a viable street stall, the going-concern value (Tobin's $Q = 83$) is real but inaccessible. She cannot sell the business: there is no title to the pitch, no enforceable transfer of the customer list, no mechanism by which a buyer can acquire the operational knowledge that keeps the firm viable. Exit means abandoning these assets entirely at zero recovery value. The rational response is to run down the firm's depreciable physical capital through use—the diary records a median depreciation of Bs 113 per week, or Bs 5,863 per year—while extracting household value through daily draws and own-consumption, until the productive assets reach the end of their useful life and the firm closes without a decision. Closure by entropy, not by choice.

These six mechanisms are not mutually exclusive; most loss-making firms exhibit several simultaneously. Their common structure is that the persistence of negative ad-

justed income is rational given the constraints each worker faces, and each constraint corresponds to a specific market failure: the labour market has not generated formal employment; savings markets have not provided accessible alternatives to firm-based liquidity; credit markets have supplied debt at usurious rates without viability assessment; property rights systems have left invisible capital non-monetisable. The informal firm is not a resilient entrepreneurial organism that survives adversity through ingenuity. It is a rational response to a set of absent markets. Interpreting its persistence as evidence of entrepreneurial dynamism—as the institutional literature surveyed in Section 2 tends to do—mistakes the symptom for a virtue.

11. Disguised Unemployment: An Accounting Definition

11.1. The Theoretical Threshold

The classical concept of disguised unemployment—workers whose marginal product falls below the subsistence wage—dates to Lewis (1954) and was formalised by Ranis and Fei (1961). Its empirical operationalisation has remained elusive, however, because identifying the relevant threshold requires knowing both what the worker earns in self-employment and what she would earn in her next-best alternative. Survey-based earnings data, as this paper documents, are too noisy to support this comparison. The diary methodology, by constructing a complete adjusted income statement, provides for the first time a clean accounting definition.

Definition 1 (Disguised Unemployment)**.** *A self-employed worker i is in* disguised unemployment *if and only if:*

$$\pi_i^{\text{adj}} < 0 \quad \textit{and} \quad \frac{\pi_i^{\text{adj}}}{h_i^o + h_i^f} < w^*, \tag{7}$$

where π_i^{adj} is adjusted net income (Definition 1), $h_i^o + h_i^f$ are total weekly labour hours, and w^* is the statutory minimum wage of Bs 13.64 per hour.

Both conditions in equation (7) are necessary. The first establishes that the firm destroys value in the economic accounting sense. The second establishes that the owner's time is priced below the market floor for unskilled labour. Together they imply that no voluntary entrepreneur—by the definitions of Schumpeter (1934), Knight (1921), or Lucas (1978)—would choose this allocation over formal wage employment at the minimum wage, absent credit constraints, skill mismatches, geographic barriers, or the absence of formal jobs altogether. The self-employment spell is therefore sustained not by entrepreneurial rents but by the non-existence of a superior outside option.

In our sample, both conditions are satisfied for all 171 firms with $\pi_i^{\text{adj}} < 0$, because the adjusted hourly return for loss-making firms is by construction negative—strictly below w^*. The 57 percent share of our sample classified as disguised unemployed under Definition 1 is not sensitive to reasonable variations in the shadow wage: at $w^* = 1.5\times$ minimum wage, the share falls modestly to 53 percent; at $w^* = 0.8\times$ minimum wage, it rises to 59 percent.

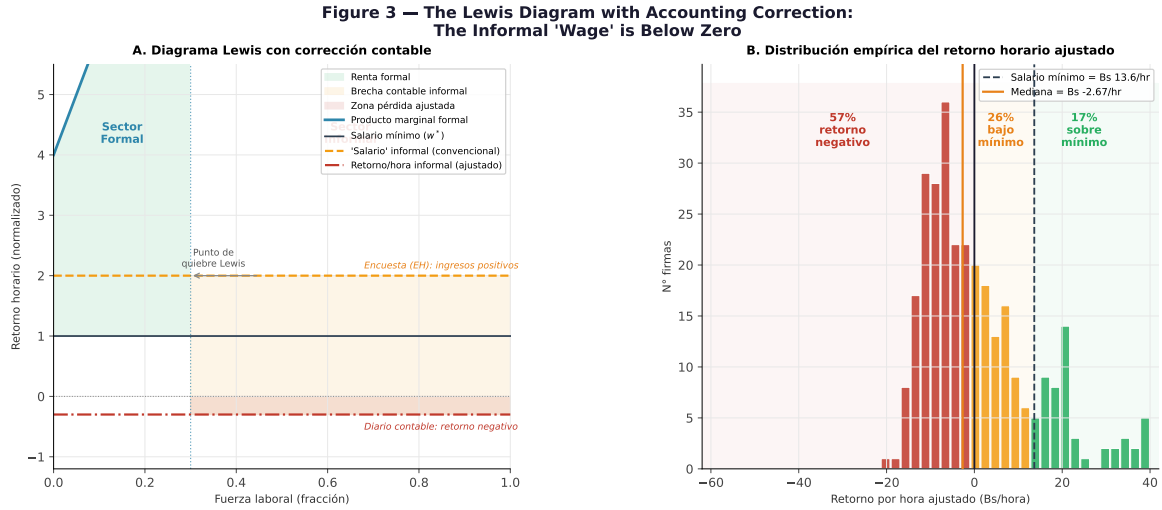

Figure 3: The Lewis Diagram with Accounting Correction. *Panel A* updates the classical Lewis (1954) dual-economy diagram. The formal sector marginal product curve (blue) intersects the statutory minimum wage (w^*, solid line). In the informal sector, survey-based accounting implies a positive subsistence "wage" (orange dashed line); diary-adjusted accounting reveals a negative hourly return (red dash-dot line). The gap between these two lines is the accounting gap made geometric: the informal sector absorbs surplus labour at returns that are not merely below the minimum wage but below zero once all input costs are correctly valued. *Panel B* shows the empirical distribution of adjusted hourly returns across all 300 firms. Red bars indicate negative returns (57% of firms); orange bars indicate returns above zero but below the minimum wage (22%); green bars indicate returns above the minimum wage (21%).

11.2. Three Types of Disguised Unemployment

Not all firms in disguised unemployment are in the same economic situation. The typological classification developed in Section 10 maps onto three structurally distinct forms of disguised unemployment, each with different policy implications.

Type A — Pure disguised unemployment (Precarious firms, $N = 147$, 49%). These firms operate with negative adjusted profit and low debt burden. They generate positive cash flow because they do not pay themselves. If a formal wage position at w^* existed in their sector, they would be better off accepting it. Their self-employment

spell is sustained by the absence of formal labour demand, not by entrepreneurial rents. They are the Lewis surplus in its purest modern form. The capital they have accumulated in tools and inventory functions as a buffer stock, not an investment: it is slowly liquidated through depreciation and household draws. Their labour market status is misclassified as "occupied self-employed" in every national dataset.

Type B — Debt-captured disguised unemployment (Critical risk firms, $N = 24$, 8%). These firms have negative adjusted profit and high debt service burdens. They cannot exit self-employment even if they wished to, because their future cash flows are pledged to informal creditors. A prestamista loan at 201 percent effective annual interest does not merely constrain investment; it eliminates the exit option. The firm must remain open to service the debt. This is disguised unemployment with a debt trap: the worker's time is not merely underpriced but has been pre-committed to creditors at usurious rates. Standard labour market interventions—job placement, skills training—cannot help this group without first addressing the debt stock.

Type C — Entrepreneurship of liquidation (subset of viable firms with high extraction, $N \approx 45$, 15%). These firms show positive adjusted profit on paper but exhibit household extraction rates above 23 percent of weekly revenue—the 75th percentile of the extraction distribution. The owner is drawing down the firm's working capital for household consumption at a rate that, if sustained, will exhaust liquidity within one to two years. They are formally classified as viable but are engaged in a process of controlled decumulation: converting the firm's balance sheet into household income, exit planned by default rather than by decision. They appear as successful informal entrepreneurs in cross-sectional data; they are not.

Only the remaining 25–30 percent of the sample—viable firms with sustainable extraction rates, positive adjusted profit, and debt burdens below 10 percent of revenue—approximate the entrepreneur of economic theory: a risk-bearer who organises production, generates surplus, and could expand with access to credit. The rest are, in varying degrees and for different structural reasons, workers for whom the informal firm is a substitute for a formal employment contract that the economy fails to provide.

11.3. Three Devastating Implications for the Literature

This taxonomy carries three implications for the received wisdom on informal-sector economics. We present them here with the bluntness they deserve.

11.3.1. Return-to-Capital Estimates Are Severely Selection-Biased

The canonical empirical literature on microenterprise development rests on a foundational claim: informal-sector entrepreneurs are capital-constrained, and relaxing that

constraint—through grants, loans, or asset transfers—generates large returns. This claim derives primarily from a series of influential randomised experiments: De Mel et al. (2008) in Sri Lanka find monthly returns of 5–7 percent on capital grants to male microentrepreneurs; McKenzie and Woodruff (2008) find returns of 20–33 percent per month in Mexico; Fafchamps et al. (2014) document high but gender-differentiated returns in Ghana. The policy implication drawn from this literature is nearly universal in development economics: capital is scarce and valuable in the informal sector, and interventions that increase capital access should be a priority.

Our data reveal why this inference is wrong for the majority of informal firms. The experimental literature estimates an average treatment effect over the full population of self-employed workers—but that population, as we document, consists of at least two economically distinct groups: the 25–30 percent who are genuinely entrepreneurial (viable firms, positive adjusted profit, sustainable extraction) and the 57 percent who are in disguised unemployment (negative adjusted profit, below-minimum-wage returns). These two groups have fundamentally different production functions for capital. For the entrepreneurial minority, capital is indeed a binding constraint: their adjusted returns on assets are positive (median $\text{ROA}^{\text{adj}} = +49.9$ percent for transport, $+27.8$ percent for manufacturing) and they are plausibly on a convex portion of a production function where the marginal product of capital exceeds the hurdle rate. For the disguised unemployed majority, capital is not the binding constraint. Their adjusted net income is negative not because they lack working capital but because they lack formal employment opportunities. Injecting capital into a food service firm whose adjusted return is -188.8 percent does not solve the problem; it extends it. The owner has more ingredients to cook with, but she still earns less than the minimum wage per hour and the imputed labour cost still swamps the gross profit.

The high average treatment effects documented in the literature are therefore consistent with our findings under one condition: the treatment effect for the entrepreneurial minority is large enough to dominate the near-zero or negative effect for the disguised unemployed majority in the weighted average. This is not a new concern in principle—Banerjee and Duflo (2007) acknowledge heterogeneous returns—but the accounting framework developed here provides the first direct evidence that the heterogeneity is not merely statistical noise around a positive mean but a structural bimodality between two qualitatively different types of workers. The policy implication is not "capital interventions do not work"; it is "capital interventions work for the 25–30 percent who are actual entrepreneurs, and the field needs an instrument to identify them before intervention rather than after." The diary methodology provides that instrument.

11.3.2. Microfinance Is Lending to Disguised Unemployment

The microfinance revolution, which began with the conviction that poor people are creditworthy entrepreneurs (Yunus, 1997), has evolved into a mature industry deploying tens of billions of dollars annually to self-employed workers in developing economies. The model rests on an assumption: the borrower is an entrepreneur whose productive opportunities are constrained by lack of credit. Our data challenge this assumption for the majority of the Bolivian informal sector, and we suspect the finding generalises.

If 57 percent of the informal self-employed are in disguised unemployment by Definition 1, and if credit is extended to informal firms without screening for entrepreneurial viability—which is the standard practice of both formal microfinance institutions and informal moneylenders—then a substantial fraction of the microcredit portfolio is directed at workers for whom borrowing makes the welfare situation worse, not better. The arithmetic is transparent: a Type A disguised unemployed worker (adjusted income −Bs 194/week before debt service) borrows Bs 3,000 from a prestamista at 10 percent monthly interest. Her weekly interest burden is Bs 69.2. Her adjusted post-debt income is −Bs 263/week. She is now further below the minimum wage floor than before the loan. The loan has not financed investment; it has financed consumption—or more precisely, it has transferred the right to the worker's future labour time, priced below the minimum wage, to the informal creditor.

The formal microcredit literature, by focusing on IFD lenders at 1.5–2.5 percent monthly rates, has documented the least harmful version of this process. Our diary data record the full distribution: 46 percent of firms are indebted, the median prestamista rate is 10 percent monthly (201 percent effective annual), and the EH survey detects only 3 percent of this debt—a detection rate of 7 percent. The Bs 46,280 in weekly debt service payments recorded across 300 firms (Table 2) is essentially invisible to the national measurement system. When we compute the welfare cost of informal credit access to disguised unemployed workers—the reduction in adjusted net income attributable to interest payments, relative to the counterfactual of no debt—we find a median weekly welfare loss of Bs 69.2 per indebted Type A worker, or Bs 3,598 per year. This is not poverty alleviation. It is the extraction of rents from the most economically vulnerable participants in the labour market, conducted by lenders who face no regulatory oversight, no interest rate ceiling, and no requirement to assess the borrower's ability to service the debt from productive rather than subsistence income.

The policy implication is not that microcredit should be abolished. It is that microcredit extended without a viability screen—without the kind of accounting assessment that the diary methodology provides—is likely to harm the majority of its borrowers in the informal sector. The first step toward a better credit policy is a better

measurement system.

11.3.3. High Informality Is Labour Market Failure, Not Entrepreneurial Dynamism

The dominant framing in development policy treats informality as a regulatory problem. The self-employed worker is informal because regulation is burdensome, formalisation costs are high, tax rates are punitive, and labour market institutions are rigid (La Porta and Shleifer, 2014). Remove these frictions, the argument goes, and entrepreneurs will formalise, productivity will increase, and growth will follow. Bolivia has pursued this logic through a succession of simplification programmes, special tax regimes for small firms (*Régimen Simplificado*, *Régimen Tributario Integrado*), and microenterprise development programmes. The informality rate has not fallen.

Our accounting evidence suggests why: the premise is wrong. The premise assumes that the informal self-employed are entrepreneurs who choose informality because its regulatory cost is lower than its benefit. Our data show that 57 percent of them are not entrepreneurs in any relevant sense. They are workers without formal employment options who have organised their own subsistence. For a Type A disguised unemployed worker in food service—a woman cooking meals at a street stall for 68 hours per week, earning Bs −915 in adjusted weekly income—the formalisation decision is irrelevant. She is not informal because formality is expensive. She is informal because there is no formal job that values her skills at the minimum wage. Formalisation cannot solve a problem that is not a regulatory problem.

The Lewis (1954) surplus labour model, in its original form, predicted that as the modern sector expanded, it would absorb the surplus workers from the traditional sector, raising wages and eliminating subsistence self-employment. Bolivia's experience suggests this process has been incomplete: the modern sector has not expanded fast enough, and the surplus workers have not been absorbed. They have instead organised themselves into an enormous informal sector that, when measured correctly, reveals the full extent of the absorption failure. Every informal firm in disguised unemployment is, by accounting identity, evidence of a formal job that the economy has not created. The 171 firms in disguised unemployment in our sample of 300 correspond, if the sample is representative, to hundreds of thousands of workers nationwide whose labour time is being misallocated to loss-making self-employment because no better option exists.

The appropriate policy response to this diagnosis is not another formalisation programme. It is investment climate reform, industrial policy, and labour market development designed to increase formal sector employment capacity. This is not a novel prescription—it is the core of structuralist development economics from Lewis (1954) through Prebisch (1950) to Rodrik (2016). What is novel is the accounting evidence:

we can now compute, at the firm level, the gap between what the worker earns in disguised unemployment and what she would earn at the minimum wage, aggregate it over the informal sector, and produce a number for the welfare cost of formal employment underabsorption. In our sample, that gap is Bs 944 per week at the median for the 171 disguised unemployed—Bs 49,088 per year per worker, at prices that are already near the poverty line. The informal sector is not a vibrant entrepreneurial ecosystem. It is, for the majority of its participants, an involuntary holding pool for the workers the formal economy has failed to employ.

12. Implications for the Measurement of Informal-Sector Income

12.1. The Dark Matter in National Accounts

The results in this paper have a direct and quantifiable implication for the measurement of informal-sector income in national accounts. Bolivia's System of National Accounts (SNA) imputes informal-sector value added using EH-reported net income as the primary source, supplemented by commodity-flow methods for sectors where output is more observable. If EH-reported net income overstates true adjusted profit by a factor of 2.7 at the median—as we document in Section 9—then the SNA's informal-sector value added is correspondingly overstated. The magnitude of the correction is large.

Consider the imputed labour component alone. Across our 300 firms, the diary records Bs 948,650 in weekly imputed labour costs (owner and family time valued at the minimum wage). This labour is real—it is physically observed, hour by hour, by trained enumerators. It is not in the SNA. It is not in the EH. It exists nowhere in Bolivia's measurement system. It is statistical dark matter: real economic activity, generating real output, performed by real workers, that has never been recorded in any account.

Scaling from the sample to the population requires care. Bolivia's INE estimates approximately 1.2 million urban informal self-employed workers in the six sectors covered by our study. If the sample is representative—by design, via stratified cluster sampling—the implied aggregate weekly imputed labour is:

$$L^{\text{agg}} = \frac{948{,}650}{300} \times 1{,}200{,}000 = \text{Bs}\,3{,}794{,}600{,}000 \text{ per week}, \tag{8}$$

or approximately Bs 197 billion per year. At 2025 prices, this is equivalent to roughly 12 percent of Bolivia's reported GDP. This is not an income transfer or a redistribution; it is output that exists—meals cooked, goods sold, passengers transported—and is

not counted in GDP because the labour input that produces it is invisible to the measurement system.

We emphasise three qualifications. First, equation (8) assumes the median imputed labour cost is representative of the informal sector as a whole; if the sectors in our sample are more labour-intensive than the full informal sector, the estimate overstates the aggregate. Second, some of the imputed labour corresponds to output already captured in the commodity-flow adjustment within the SNA; to the extent this is the case, the imputed labour is not entirely missing from GDP but merely misattributed (as capital rather than labour income). Third, the imputed labour does not represent a payment that workers receive—it represents the economic cost of their time that they do not charge themselves. The SNA correction would increase measured GDP by recording this imputed cost as both a production expense and a form of self-provided labour income, leaving household disposable income unchanged.

12.2. The Poverty Measurement Consequence

The poverty measurement consequence of the bias is more direct and more immediately actionable than the GDP consequence. ECLAC's poverty methodology (ECLAC, 2023) assigns informal self-employed workers a per-capita income based on survey-declared net income divided by household size. If survey-declared net income is Bs 31,750 per year (the EH median in our sample), and the household has three members, the per-capita income is Bs 10,583 per year—above the moderate poverty line in Bolivia. If adjusted net income is Bs −9,332 per year (the diary median in our sample), the household is unambiguously in poverty. The classification is reversed by the adjustment.

Bolivia's poverty headcount for the urban informal self-employed under ECLAC methodology is approximately 18 percent. Under adjusted accounting, the majority of these workers (57 percent, by our estimates) are loss-making, and their true economic income—after correctly valuing their own time at the minimum wage—is negative. The adjusted poverty headcount would be substantially higher. We cannot compute the exact aggregate figure without a full distributional simulation that is beyond the scope of this paper, but the direction and approximate magnitude are clear: the official poverty statistics for Bolivia's urban informal sector undercount the poor by a factor that is plausibly large.

12.3. The Survey Instrument Gap as a Policy-Actionable Scorecard

The adequacy scorecard in Section 9 (13 of 24 financial concepts absent from the EH) is not merely a methodological finding. It is an actionable agenda for the redesign of

Bolivia's household survey. The four critical absences—owner labour, family labour, informal interest, and household-firm boundary flows—can each be addressed with a targeted module addition that adds approximately 8–12 questions to the existing self-employment section. The additional interviewer time required is approximately 15 minutes per respondent. The welfare consequence of the correction—correctly classifying 57 percent of informal self-employed workers—is enormous relative to this cost.

The same scorecard applies, with minor variation, to every LSMS-type survey in the developing world. The ILO's harmonised household survey guidelines, the World Bank's LSMS protocol, and the ECLAC-UNDP MECOVI standard all share the same structural gap: they were designed for wage workers and adapted to the self-employed without the accounting discipline necessary to capture non-cash costs. The result is a global measurement system that systematically overstates the income and understates the poverty of the self-employed. Our scorecard provides the specific list of missing concepts and their priority ranking for any statistical agency seeking to correct this gap.

13. Policy Implications

The findings of this paper reorganise the policy agenda for the informal sector around a single distinction: the 43 percent of genuinely entrepreneurial firms (viable, positive adjusted profit, sustainable extraction) and the 57 percent in disguised unemployment (loss-making, working below the minimum wage, substituting self-employment for a formal job that does not exist). Every current policy instrument—microfinance, business training, formalisation programmes, special tax regimes—is designed as if the entire informal sector belongs to the first group. Our evidence shows it is a minority.

For the 43 percent entrepreneurial minority. These firms have genuine binding constraints: capital access, formalization costs, and invisible assets that cannot be collateralized. The going-concern valuation finding—Tobin's Q of 83—implies that formalisation unlocks a large option value: the invisible capital (location rents, client relationships, operational know-how) becomes legible to credit markets and collateralisable. Business training and IFD credit at 25 percent effective annual rates are welfare-improving for this group. The World Bank's *Doing Business* reforms and the IDB's formalisation agenda are correctly targeted at this minority. The problem is not the content of the policy; it is the audience.

For the 57 percent disguised unemployed. The binding constraint for this group is not capital, not regulation, and not a business registration. It is the absence of formal employment. A Type A disguised-unemployed food service worker cooking

68 hours per week for an adjusted net income of Bs −915 will not be helped by a microloan, a simplified tax regime, or a business registration. She needs a formal job at the minimum wage. The welfare gap is Bs 944 per week at the median for the 171 disguised unemployed in our sample—Bs 49,088 per year per worker at the minimum wage equivalent. Aggregated over Bolivia's urban informal sector, this is a labour market failure of first-order magnitude.

The classification problem and its solution. The central operational problem is that current measurement systems cannot distinguish these two groups. The household survey classifies all self-employed workers as profitable entrepreneurs. Under this classification, every policy dollar spent on the disguised unemployed through microcredit, business training, or formalisation support is a dollar misallocated. Our diary methodology provides the tool to make the distinction: a seven-day diary with nine modules and a daily cash reconciliation produces the adjusted income statement that determines whether the worker is genuinely entrepreneurial or disguised unemployed. At an administration cost of approximately Bs 400 per firm (one enumerator-week), this is the cheapest viability screen available. Deploying it at credit origination, transfer eligibility assessment, or policy targeting would transform the efficiency of informal sector intervention.

On taxation. The IVA (13 percent sales tax) applied to the distribution of adjusted informal income produces a cascade that Figure 4 documents. The 57 percent already in disguised unemployment pay IVA on revenue from which they are already earning negative adjusted returns; the tax converts disguised unemployment into open unemployment without creating formal jobs to absorb the displaced workers. The 43 percent entrepreneurial minority face a tax burden that reduces going-concern value but does not eliminate viability for most. The regressive incidence is not a standard public finance result—it is more severe: the tax falls on workers whose adjusted income is already negative, which means it is a transfer from below-minimum-wage workers to the public treasury, with no offsetting productivity gain.

The measurement investment. The deepest policy implication of this paper is that the measurement infrastructure underlying the informal sector policy agenda is inadequate and its inadequacy has welfare consequences. Redesigning Bolivia's EH self-employment module to capture the four critical missing concepts (owner labour, family labour, informal interest, household-firm boundary flows) would cost a small fraction of the resources currently allocated to microcredit and formalisation programmes whose targeting is corrupted by the measurement error this paper documents. The same investment, applied to the harmonised survey platforms of the ILO, World Bank, and ECLAC, would correct a global measurement failure that has been misguiding devel-

opment policy for three decades.

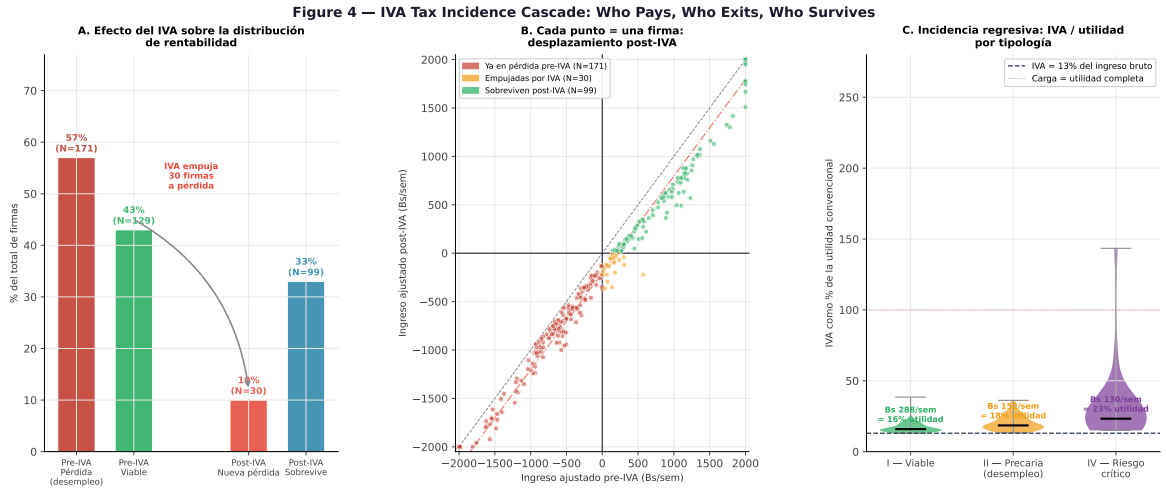

Figure 4: IVA Tax Incidence Cascade (13% sales tax). *Panel A* counts firms by status before and after applying the IVA: 171 firms (57%) were already loss-making pre-tax; the IVA pushes an additional group of previously viable firms into loss, leaving a reduced viable residual. *Panel B* plots each firm's pre-IVA vs post-IVA adjusted net income; firms below the 45-degree line are pushed down by their IVA liability, and those crossing the zero axis are newly loss-making. *Panel C* shows the regressive incidence: for Viable firms (Type I) the IVA represents a modest share of conventional profit; for Precaria firms (Type II) the IVA burden relative to an already-negative profit base is economically unbounded, meaning the tax extracts from workers already earning below the minimum wage. The IVA does not reallocate labour to higher-productivity uses—it converts disguised unemployment into open unemployment.

14. Conclusion

We have presented the first comprehensive application of double-entry financial accounting to a large-scale diary study of informal firms. The central finding is not a model-dependent prediction but an arithmetic consequence of the data: when owner and family labour are valued at the statutory minimum wage, 57 percent of the self-employed in our sample are loss-making. They are not entrepreneurs. By the accounting definition developed in Section 11, they are disguised unemployed workers whose self-employment spell substitutes for a formal wage position that the labour market fails to provide. The remaining 43 percent split into a genuinely viable entrepreneurial minority (25–30 percent) and a larger group engaged in the controlled liquidation of accumulated assets through household extraction.

This result, combined with the three implications developed in Section 11, forces a re-reading of three decades of empirical work on informal firms. The high returns

to capital documented in randomised experiments are consistent with our findings only if the treatment effect for the entrepreneurial minority dominates the effect for the disguised-unemployed majority—a compositional argument that has never been examined in that literature because the two groups have never been distinguished with accounting precision. The microfinance model's claim to poverty alleviation through credit access is undermined if its primary clients are workers earning negative returns on their time: lending at 201 percent effective annual interest to a worker already in disguised unemployment does not alleviate poverty, it monetises it. The policy narrative that high informality reflects suppressed entrepreneurship is inverted: it reflects insufficient formal employment demand, and the appropriate response is not deregulation of the informal sector but development of the formal one.

The paper's measurement contribution is the instrument that makes these inferences possible. The nine-module financial diary, documented in Appendix A, imposes double-entry discipline on every transaction across seven days, generating the income statement, balance sheet, and cash flow statement from which all results in this paper are derived. No existing survey instrument—including Bolivia's EH, the World Bank's LSMS, or any standard enterprise survey—captures the financial complexity revealed in 14,902 transactions across 300 firms. The adequacy scorecard (Section 9) shows that 13 of 24 financial concepts necessary for a complete firm account are absent from the EH. The four critical absences—owner labour, family labour, informal interest, and household-firm boundary flows—are precisely the items that determine whether a firm is profitable or in disguised unemployment. A survey instrument that cannot distinguish these two states is not a measurement tool for the informal economy. It is a mechanism for producing the illusion of one.

A. The Financial Diary Instrument

Overview and Administration Protocol

The instrument is administered over nine consecutive days: Day 0 (baseline inventory), Days 1–7 (daily transaction diary), and Day 8 (closing inventory and reconciliation interview). Each daily visit takes 25–45 minutes. All modules are administered verbally by a trained enumerator who records responses in a structured tablet form; no financial literacy is required of the respondent. The instrument is available in Spanish and Quechua. All monetary amounts are recorded in bolivianos; USD amounts are additionally recorded in the original currency with the prevailing P2P exchange rate noted at time of transaction.

Module 0 — Opening Balance Sheet (Day 0)

A. Cash and liquid assets

1. How much cash do you have *in the business* right now (separate from household money)? Bs _____
2. Do you have any business funds in a savings account, cooperativa, or mobile wallet? If yes: institution _____, balance Bs _____
3. Does anyone owe you money for goods or services sold but not yet paid? If yes, for each debtor: name/description _____, amount owed Bs _____, weeks outstanding _____

B. Inventory

cturing/food service We will count your stock together now. For each item: description _____, quantity _____, unit purchase price Bs _____, total value Bs _____

4. Of the inventory items, which are perishable (will spoil within 3 days)? Mark: _____

C. Productive assets

5. List all tools, equipment, vehicles, or fixtures used in the business. For each: description _____, year acquired _____, purchase price Bs _____, current condition (good/fair/poor) _____, estimated years of useful life remaining _____
6. Do you use any space (room, stall, vehicle, market pitch) for the business? If shared with household: what fraction is business use? _____ %

D. Liabilities

7. Do you owe money to any supplier for inputs already received but not yet paid? For each: supplier _____, amount Bs _____, due date _____
8. Do you have any outstanding loans? [*If respondent says no, probe:*] Any loan from a bank, cooperative, NGO, moneylender, family member, neighbour, *pasanaku*, or credit from a supplier for more than one month? For each: lender type _____, original amount Bs _____, amount still owed Bs _____, monthly payment Bs _____, interest rate per month % _____

E. Labour baseline

9. How many hours per day do you typically work in the business? _____ hrs
10. Does any family member (not paid a wage) regularly help? For each: relation _____, hours/day _____, days/week _____
11. If you were not running this business, what is the minimum daily wage you would accept to work for someone else doing similar work? Bs _____ / day

Module 1A — Daily Sales: Cash and Immediate Payment (Days 1–7)

[Enumerator: ask at close of each business day. Record every transaction separately, no matter how small.]

For each sale made today:

1. What did you sell? Description/item: _____
2. Quantity: _____ Unit: _____
3. Price per unit: Bs _____ (or in USD: $ _____ at rate Bs _____)
4. Total received: Bs _____
5. Payment: ☐ Cash ☐ Mobile transfer ☐ Credit (go to Module 1B)
6. Was this sale to: ☐ Individual customer ☐ Business/wholesaler ☐ Household member

Module 1B — Credit Sales and Collections (Days 1–7)

[Complete for any sale where full payment was not received today, and for any collection on a previous credit sale.]

New credit sale:

1. Item and amount as in Module 1A. Amount *not* received today: Bs _____
2. Agreed payment date: _____ Any interest agreed? ☐ No ☐ Yes: % _____

Collection on prior credit sale:

3. Amount collected today: Bs _____
4. Of this, how much is interest? Bs _____
5. Balance still owed: Bs _____

Module 1C — Own-Consumption (Days 1–7)

[For any goods produced or stocked by the business that were consumed by the household today.]

1. Item consumed: _____ Quantity: _____
2. What would you have charged a customer for this? Bs _____
3. What did it cost you to produce or purchase? Bs _____

Module 2 — Input Purchases and Supplier Payments (Days 1–7)

For each purchase of raw materials, merchandise, or ingredients today:

1. Item purchased: _____ Quantity: _____
2. Unit cost: Bs _____ (or USD: $ _____)
3. Total paid: Bs _____
4. Payment method: ☐ Cash paid today ☐ Mobile transfer ☐ Credit (pay later)
5. If credit: agreed payment date _____, any interest? _____
6. Supplier type: ☐ Market/wholesale ☐ Direct producer ☐ Import/USD supplier ☐ Other

Module 3 — Operating Expenses (Days 1–7)

[Record any business expense other than input purchases. Include only expenses paid today or accrued today.]

1. Expense category: ☐ Transport ☐ Rent/stall fee ☐ Electricity/gas/water ☐ Phone/internet ☐ Guild/gremio dues ☐ Formal wages ☐ Tax/permit ☐ Repairs ☐ Packaging ☐ Other: _____
2. Description: _____ Amount: Bs _____
3. Is this expense shared with the household? ☐ No ☐ Yes: business fraction _____ %
4. Paid: ☐ Cash ☐ Transfer ☐ Pending

Module 4A — Loans Received (Days 1–7)

[Complete if any loan disbursement was received today.]

1. Lender: ☐ Bank ☐ IFD/NGO ☐ Cooperativa ☐ Moneylender (*prestamista*) ☐ Family ☐ Neighbour/friend ☐ Supplier advance ☐ *Pasanaku*
2. Amount received: Bs _____ in USD? $ _____
3. Stated interest rate: _____ % per ☐ day ☐ week ☐ month ☐ year
4. First payment due: _____ Payment amount: Bs _____
5. Collateral or guarantee required? _____
6. Purpose of loan: ☐ Input purchase ☐ Equipment ☐ Pay another debt ☐ Household need ☐ Other: _____

Module 4B — Debt Service Payments (Days 1–7)

[Complete for every payment made on any existing debt obligation.]

1. Lender (same classification as 4A): _____
2. Total amount paid today: Bs _____
3. Of this, how much is interest? Bs _____ How much is principal repayment? Bs _____
4. Balance still owed after today's payment: Bs _____
5. Are you current on payments, or are you behind? ☐ Current ☐ Behind by _____ weeks

Module 5 — Labour (Days 1–7)

Owner hours:

1. What time did you start working in the business today? _____ Finish? _____
2. Did you take breaks? Total break time: _____ hrs
3. Net hours worked: _____ hrs
4. Main activities today (check all that apply): ☐ Selling ☐ Producing/cooking ☐ Buying inputs ☐ Transport ☐ Cleaning/prep ☐ Accounts ☐ Waiting/open ☐ Other: _____

Family workers (unpaid):

5. Did any family member help today without receiving a wage? For each:
   - Relation: _____ Age: _____ Hours worked: _____
   - Main task: _____
   - Could you have hired someone else to do this work? ☐ Yes ☐ No
   - If yes: what would you have paid them? Bs _____ / day

Paid workers:

6. Did you pay any worker (not family) today? For each: hours _____, daily wage Bs _____, task _____

Shadow wage elicitation (Day 0 only):

7. Suppose someone offered you a job doing work similar to what you do in your business. What is the minimum daily wage you would accept to stop running your business and take the job? Bs _____ / day

2 (minimum wage): Would you accept Bs 68.2 per day? ☐ Yes ☐ No (use stated amount)

Module 6 — Asset Transactions (Days 1–7)

[Complete only if an asset was acquired, disposed of, or significantly repaired today.]

1. Transaction type: ☐ Purchase ☐ Sale ☐ Repair ☐ Loss/theft ☐ Gift received
2. Asset description: _____ Amount paid/received: Bs _____
3. If purchase: financed by ☐ Cash ☐ Credit (details in Module 4A) ☐ Trade-in
4. Estimated useful life: _____ years

Module 7 — Household–Firm Boundary Flows (Days 1–7)

[Record every transfer of money or goods between the business and the household. This is the most commonly missed module; probe carefully.]

Flows from firm to household (draws):

1. Did you take any cash from the business for household expenses today? Amount: Bs _____
2. Did you use any business inputs or products for household consumption today? (Already recorded in Module 1C if yes.)
3. Did you pay any household bill (rent, utilities, school fees, food) from business funds? Amount: Bs _____, category: _____

Flows from household to firm (injections):

4. Did you put any household money into the business today to cover expenses or purchase inputs? Amount: Bs _____
5. Did a family member contribute labour to the business today that they were not compensated for? (Recorded in Module 5.)
6. Did a family member donate or lend goods to the business today? Description: _____, value: Bs _____

Module 8 — Daily Cash Reconciliation (Days 1–7)

[Complete at end of each visit. Expected closing balance = opening cash + all cash inflows − all cash outflows. If discrepancy > 5% of daily turnover, re-interview before leaving.]

1. Cash in business at opening of day: Bs _____
2. Cash inflows today (sum of M1A cash + M1B collections + M4A disbursements + M7 injections): Bs _____
3. Cash outflows today (sum of M2 cash purchases + M3 cash expenses + M4B payments + M7 draws): Bs _____
4. Expected closing balance: Bs _____
5. Actual cash on hand now (count with respondent): Bs _____
6. Discrepancy: Bs _____ Percentage of turnover: _____ %
7. If discrepancy > 5%: enumerator note on resolution: _____

Module 9 — Enumerator Qualitative Notes (Days 1–7)

1. Was today a typical business day? ☐ Yes ☐ No. If no, reason: ☐ Holiday/feria ☐ Illness ☐ Weather ☐ Supply disruption ☐ Family event ☐ Other: _____
2. Did the respondent appear to omit any transactions? Enumerator assessment (1=none omitted, 5=many likely omitted): _____. Notes: _____
3. Unusual events affecting transactions today: _____
4. Reliability of cash count (1=independently verified, 5=respondent refused): _____
5. Any concerns about respondent wellbeing, safety, or distress: _____

Closing Inventory (Day 8)

Repeat Module 0 items B (inventory) and C (assets) with physical count. Computed inventory change = Day 0 inventory − Day 8 inventory = net consumption during the week (equal to COGS plus own-consumption if no restocking). Discrepancy between computed and Module 2 purchases triggers re-interview on inventory movements.

Instrument design notes: The sequencing of modules within each day's visit follows the cash reconciliation logic: revenue modules (1A–1C) are completed first to establish inflows; expenditure modules (2, 3) establish outflows; financing modules (4A–4B) handle non-operating flows; labour module (5) records non-cash inputs; household module (7) captures boundary flows; reconciliation (8) closes the accounts. This ordering minimises recall error by anchoring the respondent's memory to physical cash movements before moving to imputed and non-cash items. Total instrument length: approximately 38 questions per day for an active trading day, 12 questions for an inactive day.

References

Ackerberg, D., Caves, K., & Frazer, G. (2015). Identification properties of recent production function estimators. *Econometrica*, 83(6), 2411–2451.

Banerjee, A. V., & Duflo, E. (2007). The economic lives of the poor. *Journal of Economic Perspectives*, 21(1), 141–167.

Banerjee, A. V., Breza, E., Duflo, E., & Kinnan, C. (2019). Can microfinance unlock a poverty trap for some entrepreneurs? *NBER Working Paper* 26346.

Berge, L. I. O., Bjorvatn, K., & Tungodden, B. (2015). Human and financial capital for microenterprise development: Evidence from a field and lab experiment. *Management Science*, 61(4), 707–722.

Bound, J., Brown, C., & Mathiowetz, N. (2001). Measurement error in survey data. *Handbook of Econometrics*, 5, 3705–3843.

Collins, D., Morduch, J., Rutherford, S., & Ruthven, O. (2009). *Portfolios of the Poor: How the World's Poor Live on $2 a Day.* Princeton University Press.

Daniels, L., & Mead, D. (1998). The contribution of small enterprises to household and national income in Kenya. *Economic Development and Cultural Change*, 47(1), 45–71.

De Mel, S., McKenzie, D., & Woodruff, C. (2008). Returns to capital in microenterprises: Evidence from a field experiment. *Quarterly Journal of Economics*, 123(4), 1329–1372.

Dupas, P., & Robinson, J. (2013). Savings constraints and microenterprise development: Evidence from a field experiment in Kenya. *American Economic Journal: Applied Economics*, 5(1), 163–192.

Fafchamps, M., McKenzie, D., Quinn, S., & Woodruff, C. (2014). Microenterprise growth and the flypaper effect: Evidence from a randomized experiment in Ghana. *Journal of Development Economics*, 106, 211–226.

Feige, E. L. (1990). Defining and estimating underground and informal economies. *World Development*, 18(7), 989–1002.

Field, E., Pande, R., Papp, J., & Rigol, N. (2013). Does the classic microfinance model discourage entrepreneurship among the poor? *American Economic Review*, 103(6), 2196–2226.

Griliches, Z., & Mairesse, J. (1995). Production functions: The search for identification. *NBER Working Paper* 5067.

Hsieh, C.-T., & Klenow, P. J. (2009). Misallocation and manufacturing TFP in China and India. *Quarterly Journal of Economics*, 124(4), 1403–1448.

International Labour Organization. (2023). *World Employment and Social Outlook: Informal Economy.* ILO.

Economic Commission for Latin America and the Caribbean. (2023). *Social Panorama of Latin America and the Caribbean 2023*. ECLAC/CEPAL.

Inter-American Development Bank. (2014). *Rethinking Productive Development: Sound Policies and Institutions for Economic Transformation*. IDB.

Inter-American Development Bank. (2020). *The Future of Work in Latin America and the Caribbean: Labour Markets in the Digital Age*. IDB.

International Labour Organization. (2018). *Women and Men in the Informal Economy: A Statistical Picture*. ILO, Geneva.

Perry, G., Maloney, W., Arias, O., Fajnzylber, P., Mason, A., & Saavedra-Chanduvi, J. (2007). *Informality: Exit and Exclusion*. World Bank Latin American and Caribbean Studies.

Prebisch, R. (1950). *The Economic Development of Latin America and its Principal Problems*. ECLAC.

Rodrik, D. (2016). Premature deindustrialization. *Journal of Economic Growth*, 21(1), 1–33.

World Bank. (2019). *World Development Report 2019: The Changing Nature of Work*. World Bank.

Yunus, M. (1997). *Banker to the Poor: Micro-lending and the Battle Against World Poverty*. PublicAffairs.

Knight, F. H. (1921). *Risk, Uncertainty and Profit*. Houghton Mifflin.

Lewis, W. A. (1954). Economic development with unlimited supplies of labour. *The Manchester School*, 22(2), 139–191.

Lucas, R. E. (1978). On the size distribution of business firms. *Bell Journal of Economics*, 9(2), 508–523.

Maloney, W. F. (2004). Informality revisited. *World Development*, 32(7), 1159–1178.

Ranis, G., & Fei, J. C. H. (1961). A theory of economic development. *American Economic Review*, 51(4), 533–565.

Schumpeter, J. A. (1934). *The Theory of Economic Development*. Harvard University Press.

Karlan, D., Kendall, J., Mann, R., Pande, R., Suri, T., & Zinman, J. (2016). Research and impacts of digital financial services. *NBER Working Paper* 22633.

Karlan, D., Ratan, A., & Zinman, J. (2014). Savings by and for the poor: A research review and agenda. *Review of Income and Wealth*, 60(1), 36–78.

La Porta, R., & Shleifer, A. (2014). Informality and development. *Journal of Economic Perspectives*, 28(3), 109–126.

Liedholm, C., & Mead, D. C. (2001). *Small Enterprises and Economic Development*. Routledge.

McKenzie, D., & Woodruff, C. (2008). Experimental evidence on returns to capital and access to finance in Mexico. *World Bank Economic Review*, 22(3), 457–482.

Meghir, C., Narita, R., & Robin, J.-M. (2015). Wages and informality in developing countries. *American Economic Review*, 105(4), 1509–1546.

Meyer, B. D., & Sullivan, J. X. (2015). Measuring the well-being of the poor using income and consumption. *Journal of Human Resources*, 38(S), 1180–1220.

Midrigan, V., & Xu, D. Y. (2014). Finance and misallocation: Evidence from plant-level data. *American Economic Review*, 104(2), 422–458.

Restuccia, D., & Rogerson, R. (2008). Policy distortions and aggregate productivity with heterogeneous establishments. *Review of Economic Dynamics*, 11(4), 707–720.

Restuccia, D., & Rogerson, R. (2017). The causes and costs of misallocation. *Journal of Economic Perspectives*, 31(3), 151–174.

Ruthven, O. (2001). Money mosaics: Financial choice and strategy in a West Delhi slum. *Journal of International Development*, 14(2), 249–271.

Schneider, F. (2002). Size and measurement of the informal economy in 110 countries around the world. *World Bank Working Paper*.

Unidad de Análisis de Políticas Sociales y Económicas. (2020). *Informe de Pobreza y Desigualdad en Bolivia*. UDAPE.

Ulyssea, G. (2018). Firms, informality, and development: Theory and evidence from Brazil. *American Economic Review*, 108(8), 2015–2047.